

VGG16_Quantization_Aware_Training_2bit4bit

December 7, 2025

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[1]: import argparse
import os
import time
import shutil

import torch
import torch.nn as nn
import torch.optim as optim
import torch.nn.functional as F
import torch.backends.cudnn as cudnn

import torchvision
import torchvision.transforms as transforms

from models import *

global best_prec
use_gpu = torch.cuda.is_available()
print('=> Building model...')

batch_size = 128
model_name = "VGG16_project_2b4b"
model = VGG16_project()

print(model)

normalize = transforms.Normalize(mean=[0.491, 0.482, 0.447], std=[0.247, 0.243, ↵
↵0.262])

train_dataset = torchvision.datasets.CIFAR10(
    root='./data',
    train=True,
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download=True,
transform=transforms.Compose([
    transforms.RandomCrop(32, padding=4),
    transforms.RandomHorizontalFlip(),
    transforms.ToTensor(),
    normalize,
]))
trainloader = torch.utils.data.DataLoader(train_dataset, batch_size=batch_size,
↳shuffle=True, num_workers=2)

test_dataset = torchvision.datasets.CIFAR10(
    root='./data',
    train=False,
    download=True,
    transform=transforms.Compose([
        transforms.ToTensor(),
        normalize,
    ]))

testloader = torch.utils.data.DataLoader(test_dataset, batch_size=batch_size,
↳shuffle=False, num_workers=2)

print_freq = 100 # every 100 batches, accuracy printed. Here, each batch
↳includes "batch_size" data points
# CIFAR10 has 50,000 training data, and 10,000 validation data.

def train(trainloader, model, criterion, optimizer, epoch):
    batch_time = AverageMeter()
    data_time = AverageMeter()
    losses = AverageMeter()
    top1 = AverageMeter()

    model.train()

    end = time.time()
    for i, (input, target) in enumerate(trainloader):
        # measure data loading time
        data_time.update(time.time() - end)

        input, target = input.cuda(), target.cuda()

        # compute output
        output = model(input)
        loss = criterion(output, target)

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        # measure accuracy and record loss
        prec = accuracy(output, target)[0]
        losses.update(loss.item(), input.size(0))
        top1.update(prec.item(), input.size(0))

        # compute gradient and do SGD step
        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        # measure elapsed time
        batch_time.update(time.time() - end)
        end = time.time()

    if i % print_freq == 0:
        print('Epoch: [{0}] [{1}/{2}]\t'
              'Time {batch_time.val:.3f} ({batch_time.avg:.3f})\t'
              'Data {data_time.val:.3f} ({data_time.avg:.3f})\t'
              'Loss {loss.val:.4f} ({loss.avg:.4f})\t'
              'Prec {top1.val:.3f}% ({top1.avg:.3f}%)'.format(
                  epoch, i, len(trainloader), batch_time=batch_time,
                  data_time=data_time, loss=losses, top1=top1))

def validate(val_loader, model, criterion ):
    batch_time = AverageMeter()
    losses = AverageMeter()
    top1 = AverageMeter()

    # switch to evaluate mode
    model.eval()

    end = time.time()
    with torch.no_grad():
        for i, (input, target) in enumerate(val_loader):

            input, target = input.cuda(), target.cuda()

            # compute output
            output = model(input)
            loss = criterion(output, target)

            # measure accuracy and record loss
            prec = accuracy(output, target)[0]
            losses.update(loss.item(), input.size(0))

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        top1.update(prec.item(), input.size(0))

        # measure elapsed time
        batch_time.update(time.time() - end)
        end = time.time()

        if i % print_freq == 0: # This line shows how frequently print out
            ↪ the status. e.g., i%5 => every 5 batch, prints out
            print('Test: [{0}/{1}]\t'
                  'Time {batch_time.val:.3f} ({batch_time.avg:.3f})\t'
                  'Loss {loss.val:.4f} ({loss.avg:.4f})\t'
                  'Prec {top1.val:.3f}% ({top1.avg:.3f}%)'.format(
                      i, len(val_loader), batch_time=batch_time, loss=losses,
                      top1=top1))

    print(' * Prec {top1.avg:.3f}% '.format(top1=top1))
    return top1.avg

def accuracy(output, target, topk=(1,)):
    """Computes the precision@k for the specified values of k"""
    maxk = max(topk)
    batch_size = target.size(0)

    _, pred = output.topk(maxk, 1, True, True)
    pred = pred.t()
    correct = pred.eq(target.view(1, -1).expand_as(pred))

    res = []
    for k in topk:
        correct_k = correct[:k].view(-1).float().sum(0)
        res.append(correct_k.mul_(100.0 / batch_size))
    return res

class AverageMeter(object):
    """Computes and stores the average and current value"""
    def __init__(self):
        self.reset()

    def reset(self):
        self.val = 0
        self.avg = 0
        self.sum = 0
        self.count = 0

    def update(self, val, n=1):

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        self.val = val
        self.sum += val * n
        self.count += n
        self.avg = self.sum / self.count

def save_checkpoint(state, is_best, fdir):
    filepath = os.path.join(fdir, 'checkpoint.pth')
    torch.save(state, filepath)
    if is_best:
        shutil.copyfile(filepath, os.path.join(fdir, 'model_best.pth.tar'))

#model = nn.DataParallel(model).cuda()
#all_params = checkpoint['state_dict']
#model.load_state_dict(all_params, strict=False)
#criterion = nn.CrossEntropyLoss().cuda()
#validate(testloader, model, criterion)

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=> Building model...

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VGG_quant(
  (features): Sequential(
    (0): QuantConv2d(
      3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
    (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU(inplace=True)
    (3): QuantConv2d(
      64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
    (4): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (5): ReLU(inplace=True)
    (6): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (7): QuantConv2d(
      64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
    (8): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (9): ReLU(inplace=True)
    (10): QuantConv2d(
      128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
      (weight_quant): weight_quantize_fn()
    )
  )
)

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    )
    (11): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (12): ReLU(inplace=True)
    (13): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (14): QuantConv2d(
        128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (15): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (16): ReLU(inplace=True)
    (17): QuantConv2d(
        256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (18): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (19): ReLU(inplace=True)
    (20): QuantConv2d(
        256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (21): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (22): ReLU(inplace=True)
    (23): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (24): QuantConv2d(
        256, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (25): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (26): ReLU(inplace=True)
    (27): QuantConv2d(
        16, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (28): ReLU(inplace=True)
    (29): QuantConv2d(
        16, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
        (weight_quant): weight_quantize_fn()
    )
    (30): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (31): ReLU(inplace=True)

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(32): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
(33): QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
)
(34): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
(35): ReLU(inplace=True)
(36): QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
)
(37): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
(38): ReLU(inplace=True)
(39): QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
)
(40): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
(41): ReLU(inplace=True)
(42): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
(43): AvgPool2d(kernel_size=1, stride=1, padding=0)
)
(classifier): Linear(in_features=512, out_features=10, bias=True)
)

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[2]: def adjust_learning_rate(optimizer, epoch):
      """For resnet, the lr starts from 0.1, and is divided by 10 at 80 and 120_
      ↪epochs"""
      adjust_list = [100, 120]
      if epoch in adjust_list:
          for param_group in optimizer.param_groups:
              param_group['lr'] = param_group['lr'] * 0.1

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[3]: # This cell won't be given, but students will complete the training

lr = 4e-2
weight_decay = 1e-4
epochs = 150
best_prec = 0

#model = nn.DataParallel(model).cuda()
model.cuda()

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criterion = nn.CrossEntropyLoss().cuda()
optimizer = torch.optim.SGD(model.parameters(), lr=lr, momentum=0.9,
    ↪weight_decay=weight_decay)
#cudnn.benchmark = True

if not os.path.exists('result'):
    os.makedirs('result')
fdir = 'result/'+str(model_name)
if not os.path.exists(fdir):
    os.makedirs(fdir)

for epoch in range(0, epochs):
    adjust_learning_rate(optimizer, epoch)

    train(trainloader, model, criterion, optimizer, epoch)

    # evaluate on test set
    print("Validation starts")
    prec = validate(testloader, model, criterion)

    # remember best precision and save checkpoint
    is_best = prec > best_prec
    best_prec = max(prec,best_prec)
    print('best acc: {:.1f}'.format(best_prec))
    save_checkpoint({
        'epoch': epoch + 1,
        'state_dict': model.state_dict(),
        'best_prec': best_prec,
        'optimizer': optimizer.state_dict(),
    }, is_best, fdir)

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Epoch: [0][0/391]      Time 0.702 (0.702)      Data 0.110 (0.110)      Loss
2.4724 (2.4724)      Prec 9.375% (9.375%)
Epoch: [0][100/391]    Time 0.035 (0.034)      Data 0.007 (0.004)      Loss
2.2994 (3.0935)      Prec 12.500% (10.791%)
Epoch: [0][200/391]    Time 0.029 (0.030)      Data 0.009 (0.005)      Loss
2.3398 (2.7141)      Prec 12.500% (11.194%)
Epoch: [0][300/391]    Time 0.028 (0.028)      Data 0.008 (0.005)      Loss
2.2924 (2.5766)      Prec 17.188% (11.698%)
Validation starts
Test: [0/79]      Time 0.096 (0.096)      Loss 2.2312 (2.2312)      Prec 14.062%
(14.062%)
* Prec 13.530%
best acc: 13.530000
Epoch: [1][0/391]      Time 0.140 (0.140)      Data 0.111 (0.111)      Loss
2.3381 (2.3381)      Prec 11.719% (11.719%)

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Epoch: [1][100/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
 2.2930 (2.2765) Prec 14.062% (14.364%)
 Epoch: [1][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
 2.0824 (2.2631) Prec 18.750% (14.782%)
 Epoch: [1][300/391] Time 0.022 (0.025) Data 0.000 (0.005) Loss
 2.1651 (2.2296) Prec 11.719% (15.677%)
 Validation starts
 Test: [0/79] Time 0.092 (0.092) Loss 2.0925 (2.0925) Prec 21.094%
 (21.094%)
 * Prec 23.450%
 best acc: 23.450000
 Epoch: [2][0/391] Time 0.143 (0.143) Data 0.111 (0.111) Loss
 2.1230 (2.1230) Prec 24.219% (24.219%)
 Epoch: [2][100/391] Time 0.031 (0.027) Data 0.011 (0.006) Loss
 1.9520 (2.0228) Prec 22.656% (23.043%)
 Epoch: [2][200/391] Time 0.030 (0.026) Data 0.011 (0.006) Loss
 2.0285 (1.9978) Prec 28.906% (23.554%)
 Epoch: [2][300/391] Time 0.030 (0.026) Data 0.010 (0.005) Loss
 1.8853 (1.9909) Prec 27.344% (23.796%)
 Validation starts
 Test: [0/79] Time 0.091 (0.091) Loss 1.9235 (1.9235) Prec 25.781%
 (25.781%)
 * Prec 27.110%
 best acc: 27.110000
 Epoch: [3][0/391] Time 0.140 (0.140) Data 0.111 (0.111) Loss
 1.9770 (1.9770) Prec 29.688% (29.688%)
 Epoch: [3][100/391] Time 0.020 (0.027) Data 0.000 (0.005) Loss
 1.8631 (1.9135) Prec 26.562% (27.529%)
 Epoch: [3][200/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
 2.0195 (1.8926) Prec 20.312% (28.409%)
 Epoch: [3][300/391] Time 0.021 (0.025) Data 0.000 (0.005) Loss
 2.0154 (1.8712) Prec 30.469% (29.573%)
 Validation starts
 Test: [0/79] Time 0.092 (0.092) Loss 1.8882 (1.8882) Prec 34.375%
 (34.375%)
 * Prec 29.970%
 best acc: 29.970000
 Epoch: [4][0/391] Time 0.141 (0.141) Data 0.112 (0.112) Loss
 1.8320 (1.8320) Prec 32.812% (32.812%)
 Epoch: [4][100/391] Time 0.029 (0.026) Data 0.010 (0.005) Loss
 1.8816 (1.7736) Prec 28.906% (33.942%)
 Epoch: [4][200/391] Time 0.029 (0.026) Data 0.009 (0.005) Loss
 1.7602 (1.7665) Prec 35.938% (34.383%)
 Epoch: [4][300/391] Time 0.029 (0.026) Data 0.009 (0.005) Loss
 1.7947 (1.7562) Prec 38.281% (34.920%)
 Validation starts
 Test: [0/79] Time 0.093 (0.093) Loss 1.6417 (1.6417) Prec 37.500%
 (37.500%)

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* Prec 37.210%
best acc: 37.210000
Epoch: [5][0/391]      Time 0.151 (0.151)      Data 0.114 (0.114)      Loss
1.6624 (1.6624)      Prec 33.594% (33.594%)
Epoch: [5][100/391]    Time 0.025 (0.027)      Data 0.005 (0.005)      Loss
1.6865 (1.6885)      Prec 35.156% (38.049%)
Epoch: [5][200/391]    Time 0.036 (0.026)      Data 0.016 (0.005)      Loss
1.5827 (1.6804)      Prec 41.406% (38.145%)
Epoch: [5][300/391]    Time 0.029 (0.026)      Data 0.009 (0.005)      Loss
1.6537 (1.6677)      Prec 42.188% (38.684%)
Validation starts
Test: [0/79]      Time 0.093 (0.093)      Loss 1.6600 (1.6600)      Prec 39.844%
(39.844%)
* Prec 40.100%
best acc: 40.100000
Epoch: [6][0/391]      Time 0.144 (0.144)      Data 0.112 (0.112)      Loss
1.6662 (1.6662)      Prec 32.812% (32.812%)
Epoch: [6][100/391]    Time 0.034 (0.026)      Data 0.014 (0.006)      Loss
1.7409 (1.6004)      Prec 38.281% (41.306%)
Epoch: [6][200/391]    Time 0.030 (0.026)      Data 0.010 (0.006)      Loss
1.6606 (1.5894)      Prec 41.406% (41.663%)
Epoch: [6][300/391]    Time 0.026 (0.025)      Data 0.007 (0.005)      Loss
1.5137 (1.5827)      Prec 46.094% (42.297%)
Validation starts
Test: [0/79]      Time 0.098 (0.098)      Loss 1.5506 (1.5506)      Prec 42.188%
(42.188%)
* Prec 44.620%
best acc: 44.620000
Epoch: [7][0/391]      Time 0.148 (0.148)      Data 0.113 (0.113)      Loss
1.4973 (1.4973)      Prec 46.875% (46.875%)
Epoch: [7][100/391]    Time 0.021 (0.027)      Data 0.002 (0.005)      Loss
1.6188 (1.5365)      Prec 38.281% (44.756%)
Epoch: [7][200/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
1.6609 (1.5251)      Prec 44.531% (44.908%)
Epoch: [7][300/391]    Time 0.020 (0.026)      Data 0.000 (0.006)      Loss
1.5752 (1.5223)      Prec 45.312% (45.206%)
Validation starts
Test: [0/79]      Time 0.099 (0.099)      Loss 1.6402 (1.6402)      Prec 45.312%
(45.312%)
* Prec 46.310%
best acc: 46.310000
Epoch: [8][0/391]      Time 0.145 (0.145)      Data 0.115 (0.115)      Loss
1.5249 (1.5249)      Prec 39.844% (39.844%)
Epoch: [8][100/391]    Time 0.027 (0.026)      Data 0.008 (0.005)      Loss
1.4071 (1.4720)      Prec 46.094% (46.658%)
Epoch: [8][200/391]    Time 0.029 (0.025)      Data 0.008 (0.005)      Loss
1.5135 (1.4661)      Prec 45.312% (47.143%)
Epoch: [8][300/391]    Time 0.020 (0.025)      Data 0.000 (0.005)      Loss

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1.5313 (1.4650) Prec 44.531% (47.288%)
Validation starts
Test: [0/79] Time 0.092 (0.092) Loss 1.6290 (1.6290) Prec 38.281% (38.281%)
* Prec 46.870%
best acc: 46.870000
Epoch: [9][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss 1.3721 (1.3721) Prec 48.438% (48.438%)
Epoch: [9][100/391] Time 0.020 (0.029) Data 0.000 (0.004) Loss 1.3387 (1.4137) Prec 56.250% (48.925%)
Epoch: [9][200/391] Time 0.020 (0.027) Data 0.000 (0.005) Loss 1.2521 (1.4129) Prec 53.125% (49.265%)
Epoch: [9][300/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss 1.4549 (1.4089) Prec 50.000% (49.286%)
Validation starts
Test: [0/79] Time 0.094 (0.094) Loss 1.5128 (1.5128) Prec 45.312% (45.312%)
* Prec 49.350%
best acc: 49.350000
Epoch: [10][0/391] Time 0.152 (0.152) Data 0.115 (0.115) Loss 1.4474 (1.4474) Prec 48.438% (48.438%)
Epoch: [10][100/391] Time 0.029 (0.026) Data 0.009 (0.005) Loss 1.2981 (1.3656) Prec 50.781% (50.982%)
Epoch: [10][200/391] Time 0.028 (0.026) Data 0.007 (0.005) Loss 1.3239 (1.3524) Prec 51.562% (51.562%)
Epoch: [10][300/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss 1.4188 (1.3447) Prec 47.656% (51.848%)
Validation starts
Test: [0/79] Time 0.095 (0.095) Loss 1.2461 (1.2461) Prec 57.031% (57.031%)
* Prec 54.300%
best acc: 54.300000
Epoch: [11][0/391] Time 0.141 (0.141) Data 0.112 (0.112) Loss 1.2518 (1.2518) Prec 57.031% (57.031%)
Epoch: [11][100/391] Time 0.027 (0.026) Data 0.008 (0.005) Loss 1.0775 (1.2913) Prec 66.406% (53.775%)
Epoch: [11][200/391] Time 0.025 (0.025) Data 0.005 (0.005) Loss 1.4054 (1.2888) Prec 50.781% (53.961%)
Epoch: [11][300/391] Time 0.028 (0.025) Data 0.009 (0.005) Loss 1.2783 (1.2879) Prec 53.906% (53.808%)
Validation starts
Test: [0/79] Time 0.098 (0.098) Loss 1.2185 (1.2185) Prec 58.594% (58.594%)
* Prec 54.800%
best acc: 54.800000
Epoch: [12][0/391] Time 0.140 (0.140) Data 0.111 (0.111) Loss 1.3487 (1.3487) Prec 52.344% (52.344%)
Epoch: [12][100/391] Time 0.030 (0.026) Data 0.008 (0.005) Loss

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1.4587 (1.2495)    Prec 53.125% (55.801%)
Epoch: [12][200/391]    Time 0.025 (0.025)    Data 0.006 (0.005)    Loss
1.0718 (1.2429)    Prec 57.812% (55.706%)
Epoch: [12][300/391]    Time 0.031 (0.025)    Data 0.011 (0.005)    Loss
1.0626 (1.2404)    Prec 60.156% (55.804%)
Validation starts
Test: [0/79]    Time 0.096 (0.096)    Loss 1.4611 (1.4611)    Prec 50.000%
(50.000%)
* Prec 52.470%
best acc: 54.800000
Epoch: [13][0/391]    Time 0.155 (0.155)    Data 0.114 (0.114)    Loss
1.3037 (1.3037)    Prec 50.781% (50.781%)
Epoch: [13][100/391]    Time 0.021 (0.026)    Data 0.002 (0.005)    Loss
1.1455 (1.1992)    Prec 57.031% (57.232%)
Epoch: [13][200/391]    Time 0.020 (0.025)    Data 0.000 (0.005)    Loss
1.0958 (1.2094)    Prec 62.500% (57.101%)
Epoch: [13][300/391]    Time 0.020 (0.026)    Data 0.000 (0.005)    Loss
1.0936 (1.2029)    Prec 63.281% (57.387%)
Validation starts
Test: [0/79]    Time 0.090 (0.090)    Loss 1.2133 (1.2133)    Prec 53.906%
(53.906%)
* Prec 55.740%
best acc: 55.740000
Epoch: [14][0/391]    Time 0.148 (0.148)    Data 0.112 (0.112)    Loss
1.1109 (1.1109)    Prec 57.812% (57.812%)
Epoch: [14][100/391]    Time 0.022 (0.028)    Data 0.000 (0.005)    Loss
1.1190 (1.1602)    Prec 58.594% (58.385%)
Epoch: [14][200/391]    Time 0.019 (0.026)    Data 0.000 (0.005)    Loss
1.1348 (1.1577)    Prec 61.719% (58.804%)
Epoch: [14][300/391]    Time 0.022 (0.025)    Data 0.002 (0.004)    Loss
1.0875 (1.1563)    Prec 63.281% (58.955%)
Validation starts
Test: [0/79]    Time 0.093 (0.093)    Loss 1.1780 (1.1780)    Prec 57.812%
(57.812%)
* Prec 59.210%
best acc: 59.210000
Epoch: [15][0/391]    Time 0.141 (0.141)    Data 0.112 (0.112)    Loss
1.1792 (1.1792)    Prec 55.469% (55.469%)
Epoch: [15][100/391]    Time 0.032 (0.026)    Data 0.012 (0.005)    Loss
1.3064 (1.1224)    Prec 53.125% (60.094%)
Epoch: [15][200/391]    Time 0.029 (0.025)    Data 0.009 (0.005)    Loss
1.0921 (1.1276)    Prec 62.500% (59.865%)
Epoch: [15][300/391]    Time 0.033 (0.025)    Data 0.013 (0.005)    Loss
1.2831 (1.1230)    Prec 53.125% (60.177%)
Validation starts
Test: [0/79]    Time 0.088 (0.088)    Loss 1.3567 (1.3567)    Prec 52.344%
(52.344%)
* Prec 55.140%

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best acc: 59.210000
Epoch: [16][0/391]      Time 0.143 (0.143)      Data 0.112 (0.112)      Loss
1.1853 (1.1853)    Prec 53.906% (53.906%)
Epoch: [16][100/391]    Time 0.025 (0.026)      Data 0.003 (0.005)      Loss
1.0813 (1.0850)    Prec 62.500% (61.680%)
Epoch: [16][200/391]    Time 0.029 (0.025)      Data 0.008 (0.005)      Loss
1.1111 (1.0827)    Prec 53.906% (61.676%)
Epoch: [16][300/391]    Time 0.029 (0.025)      Data 0.009 (0.005)      Loss
1.0129 (1.0885)    Prec 59.375% (61.579%)
Validation starts
Test: [0/79]      Time 0.092 (0.092)      Loss 1.0742 (1.0742)      Prec 61.719%
(61.719%)
* Prec 61.590%
best acc: 61.590000
Epoch: [17][0/391]      Time 0.141 (0.141)      Data 0.112 (0.112)      Loss
1.2420 (1.2420)    Prec 55.469% (55.469%)
Epoch: [17][100/391]    Time 0.032 (0.027)      Data 0.008 (0.006)      Loss
1.0859 (1.0611)    Prec 61.719% (62.570%)
Epoch: [17][200/391]    Time 0.028 (0.026)      Data 0.008 (0.005)      Loss
0.8645 (1.0562)    Prec 71.875% (62.893%)
Epoch: [17][300/391]    Time 0.024 (0.025)      Data 0.003 (0.005)      Loss
0.9423 (1.0583)    Prec 67.188% (62.949%)
Validation starts
Test: [0/79]      Time 0.092 (0.092)      Loss 1.1714 (1.1714)      Prec 62.500%
(62.500%)
* Prec 58.830%
best acc: 61.590000
Epoch: [18][0/391]      Time 0.140 (0.140)      Data 0.112 (0.112)      Loss
1.0669 (1.0669)    Prec 64.844% (64.844%)
Epoch: [18][100/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
0.9990 (1.0491)    Prec 64.062% (63.281%)
Epoch: [18][200/391]    Time 0.021 (0.025)      Data 0.002 (0.005)      Loss
1.0158 (1.0282)    Prec 64.844% (63.985%)
Epoch: [18][300/391]    Time 0.030 (0.025)      Data 0.010 (0.005)      Loss
0.9740 (1.0252)    Prec 67.969% (64.156%)
Validation starts
Test: [0/79]      Time 0.091 (0.091)      Loss 1.0698 (1.0698)      Prec 60.156%
(60.156%)
* Prec 60.440%
best acc: 61.590000
Epoch: [19][0/391]      Time 0.141 (0.141)      Data 0.112 (0.112)      Loss
1.1736 (1.1736)    Prec 58.594% (58.594%)
Epoch: [19][100/391]    Time 0.028 (0.026)      Data 0.001 (0.006)      Loss
1.0168 (0.9943)    Prec 64.844% (65.169%)
Epoch: [19][200/391]    Time 0.021 (0.026)      Data 0.000 (0.005)      Loss
1.0246 (0.9872)    Prec 64.844% (65.520%)
Epoch: [19][300/391]    Time 0.020 (0.025)      Data 0.000 (0.005)      Loss
0.9650 (0.9928)    Prec 62.500% (65.189%)

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Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 1.1102 (1.1102) Prec 62.500% (62.500%)

* Prec 61.120%

best acc: 61.590000

Epoch: [20] [0/391]	Time 0.141 (0.141)	Data 0.113 (0.113)	Loss 0.8686 (0.8686)	Prec 67.188% (67.188%)
Epoch: [20] [100/391]	Time 0.020 (0.026)	Data 0.000 (0.006)	Loss 1.0572 (0.9681)	Prec 60.156% (66.499%)
Epoch: [20] [200/391]	Time 0.020 (0.026)	Data 0.000 (0.005)	Loss 0.9465 (0.9675)	Prec 62.500% (66.422%)
Epoch: [20] [300/391]	Time 0.020 (0.025)	Data 0.000 (0.005)	Loss 0.8766 (0.9664)	Prec 67.969% (66.393%)

Validation starts

Test: [0/79] Time 0.100 (0.100) Loss 0.9385 (0.9385) Prec 67.188% (67.188%)

* Prec 64.260%

best acc: 64.260000

Epoch: [21] [0/391]	Time 0.142 (0.142)	Data 0.112 (0.112)	Loss 0.9962 (0.9962)	Prec 66.406% (66.406%)
Epoch: [21] [100/391]	Time 0.023 (0.025)	Data 0.002 (0.005)	Loss 1.0257 (0.9471)	Prec 58.594% (67.218%)
Epoch: [21] [200/391]	Time 0.020 (0.025)	Data 0.000 (0.005)	Loss 1.1451 (0.9401)	Prec 64.844% (67.483%)
Epoch: [21] [300/391]	Time 0.020 (0.025)	Data 0.000 (0.005)	Loss 1.1486 (0.9389)	Prec 55.469% (67.429%)

Validation starts

Test: [0/79] Time 0.090 (0.090) Loss 1.1076 (1.1076) Prec 59.375% (59.375%)

* Prec 62.970%

best acc: 64.260000

Epoch: [22] [0/391]	Time 0.151 (0.151)	Data 0.113 (0.113)	Loss 0.8546 (0.8546)	Prec 75.781% (75.781%)
Epoch: [22] [100/391]	Time 0.028 (0.026)	Data 0.008 (0.005)	Loss 0.7790 (0.9351)	Prec 78.125% (67.628%)
Epoch: [22] [200/391]	Time 0.020 (0.026)	Data 0.000 (0.005)	Loss 1.0725 (0.9280)	Prec 60.156% (67.969%)
Epoch: [22] [300/391]	Time 0.020 (0.026)	Data 0.000 (0.005)	Loss 1.0474 (0.9192)	Prec 65.625% (68.158%)

Validation starts

Test: [0/79] Time 0.097 (0.097) Loss 0.9369 (0.9369) Prec 67.969% (67.969%)

* Prec 66.360%

best acc: 66.360000

Epoch: [23] [0/391]	Time 0.152 (0.152)	Data 0.114 (0.114)	Loss 1.0411 (1.0411)	Prec 64.844% (64.844%)
Epoch: [23] [100/391]	Time 0.021 (0.027)	Data 0.002 (0.006)	Loss 0.9819 (0.8893)	Prec 66.406% (69.160%)

Epoch: [23][200/391] Time 0.027 (0.027) Data 0.007 (0.005) Loss
1.2754 (0.8861) Prec 53.125% (69.368%)

Epoch: [23][300/391] Time 0.030 (0.026) Data 0.010 (0.005) Loss
0.9216 (0.8848) Prec 71.094% (69.347%)

Validation starts
Test: [0/79] Time 0.092 (0.092) Loss 0.9011 (0.9011) Prec 68.750%
(68.750%)
* Prec 66.150%
best acc: 66.360000

Epoch: [24][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss
0.8772 (0.8772) Prec 71.094% (71.094%)

Epoch: [24][100/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
0.8999 (0.8806) Prec 67.969% (69.717%)

Epoch: [24][200/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
0.9149 (0.8699) Prec 69.531% (69.951%)

Epoch: [24][300/391] Time 0.030 (0.025) Data 0.010 (0.005) Loss
0.7325 (0.8707) Prec 71.094% (69.871%)

Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.8483 (0.8483) Prec 70.312%
(70.312%)
* Prec 69.300%
best acc: 69.300000

Epoch: [25][0/391] Time 0.152 (0.152) Data 0.114 (0.114) Loss
0.6924 (0.6924) Prec 77.344% (77.344%)

Epoch: [25][100/391] Time 0.028 (0.027) Data 0.008 (0.006) Loss
0.7362 (0.8477) Prec 76.562% (70.591%)

Epoch: [25][200/391] Time 0.021 (0.026) Data 0.002 (0.005) Loss
0.9198 (0.8487) Prec 67.969% (70.546%)

Epoch: [25][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.8417 (0.8485) Prec 70.312% (70.756%)

Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 1.0298 (1.0298) Prec 66.406%
(66.406%)
* Prec 65.410%
best acc: 69.300000

Epoch: [26][0/391] Time 0.144 (0.144) Data 0.115 (0.115) Loss
0.6603 (0.6603) Prec 75.000% (75.000%)

Epoch: [26][100/391] Time 0.030 (0.026) Data 0.010 (0.006) Loss
0.8912 (0.8003) Prec 71.875% (72.215%)

Epoch: [26][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.8492 (0.8188) Prec 68.750% (71.743%)

Epoch: [26][300/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.6579 (0.8208) Prec 78.906% (71.538%)

Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.8293 (0.8293) Prec 71.875%
(71.875%)
* Prec 70.480%
best acc: 70.480000

Epoch: [27][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss
 0.7541 (0.7541) Prec 72.656% (72.656%)
 Epoch: [27][100/391] Time 0.027 (0.025) Data 0.008 (0.006) Loss
 0.6078 (0.8094) Prec 78.906% (72.123%)
 Epoch: [27][200/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
 0.6240 (0.8129) Prec 79.688% (71.891%)
 Epoch: [27][300/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
 0.6743 (0.8101) Prec 77.344% (72.015%)
 Validation starts
 Test: [0/79] Time 0.090 (0.090) Loss 0.9803 (0.9803) Prec 70.312%
 (70.312%)
 * Prec 68.660%
 best acc: 70.480000
 Epoch: [28][0/391] Time 0.145 (0.145) Data 0.116 (0.116) Loss
 0.7432 (0.7432) Prec 78.906% (78.906%)
 Epoch: [28][100/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
 0.7325 (0.7938) Prec 70.312% (72.478%)
 Epoch: [28][200/391] Time 0.040 (0.026) Data 0.019 (0.005) Loss
 0.7286 (0.7930) Prec 75.781% (72.707%)
 Epoch: [28][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
 0.8630 (0.7941) Prec 71.875% (72.661%)
 Validation starts
 Test: [0/79] Time 0.093 (0.093) Loss 0.9624 (0.9624) Prec 67.188%
 (67.188%)
 * Prec 68.450%
 best acc: 70.480000
 Epoch: [29][0/391] Time 0.141 (0.141) Data 0.112 (0.112) Loss
 0.9352 (0.9352) Prec 64.844% (64.844%)
 Epoch: [29][100/391] Time 0.025 (0.026) Data 0.000 (0.006) Loss
 0.8849 (0.7665) Prec 71.875% (73.685%)
 Epoch: [29][200/391] Time 0.025 (0.025) Data 0.002 (0.005) Loss
 0.8035 (0.7738) Prec 76.562% (73.383%)
 Epoch: [29][300/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
 0.9303 (0.7727) Prec 67.969% (73.352%)
 Validation starts
 Test: [0/79] Time 0.099 (0.099) Loss 0.7671 (0.7671) Prec 75.781%
 (75.781%)
 * Prec 71.180%
 best acc: 71.180000
 Epoch: [30][0/391] Time 0.145 (0.145) Data 0.116 (0.116) Loss
 0.8215 (0.8215) Prec 71.094% (71.094%)
 Epoch: [30][100/391] Time 0.023 (0.026) Data 0.000 (0.006) Loss
 0.7572 (0.7563) Prec 71.094% (73.824%)
 Epoch: [30][200/391] Time 0.021 (0.026) Data 0.002 (0.005) Loss
 0.5944 (0.7537) Prec 79.688% (73.939%)
 Epoch: [30][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
 0.7085 (0.7566) Prec 75.781% (73.798%)
 Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.9485 (0.9485) Prec 66.406%
(66.406%)

* Prec 69.540%

best acc: 71.180000

Epoch: [31][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss
0.7525 (0.7525) Prec 77.344% (77.344%)

Epoch: [31][100/391] Time 0.022 (0.026) Data 0.002 (0.006) Loss
0.8608 (0.7366) Prec 71.094% (74.791%)

Epoch: [31][200/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.7109 (0.7408) Prec 75.000% (74.658%)

Epoch: [31][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.6910 (0.7411) Prec 75.781% (74.486%)

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.6988 (0.6988) Prec 78.906%
(78.906%)

* Prec 72.040%

best acc: 72.040000

Epoch: [32][0/391] Time 0.149 (0.149) Data 0.112 (0.112) Loss
0.8298 (0.8298) Prec 71.875% (71.875%)

Epoch: [32][100/391] Time 0.020 (0.028) Data 0.000 (0.005) Loss
0.8621 (0.7181) Prec 66.406% (75.387%)

Epoch: [32][200/391] Time 0.023 (0.027) Data 0.001 (0.005) Loss
0.7397 (0.7233) Prec 72.656% (75.334%)

Epoch: [32][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.7480 (0.7227) Prec 69.531% (75.265%)

Validation starts

Test: [0/79] Time 0.096 (0.096) Loss 0.8295 (0.8295) Prec 70.312%
(70.312%)

* Prec 71.650%

best acc: 72.040000

Epoch: [33][0/391] Time 0.152 (0.152) Data 0.114 (0.114) Loss
0.7971 (0.7971) Prec 73.438% (73.438%)

Epoch: [33][100/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
0.6996 (0.7049) Prec 78.125% (75.704%)

Epoch: [33][200/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
0.6846 (0.7168) Prec 77.344% (75.323%)

Epoch: [33][300/391] Time 0.027 (0.025) Data 0.007 (0.004) Loss
0.7557 (0.7135) Prec 73.438% (75.436%)

Validation starts

Test: [0/79] Time 0.096 (0.096) Loss 0.7548 (0.7548) Prec 73.438%
(73.438%)

* Prec 72.870%

best acc: 72.870000

Epoch: [34][0/391] Time 0.142 (0.142) Data 0.114 (0.114) Loss
0.6601 (0.6601) Prec 75.000% (75.000%)

Epoch: [34][100/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
0.6720 (0.6948) Prec 79.688% (76.006%)

Epoch: [34][200/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss

0.7150 (0.6981) Prec 73.438% (75.925%)
Epoch: [34][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.5604 (0.6987) Prec 80.469% (76.010%)
Validation starts
Test: [0/79] Time 0.093 (0.093) Loss 0.7896 (0.7896) Prec 73.438%
(73.438%)
* Prec 72.540%
best acc: 72.870000
Epoch: [35][0/391] Time 0.158 (0.158) Data 0.116 (0.116) Loss
0.5800 (0.5800) Prec 77.344% (77.344%)
Epoch: [35][100/391] Time 0.031 (0.027) Data 0.012 (0.006) Loss
0.5911 (0.6841) Prec 77.344% (76.176%)
Epoch: [35][200/391] Time 0.031 (0.026) Data 0.012 (0.006) Loss
0.7227 (0.6879) Prec 73.438% (76.325%)
Epoch: [35][300/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.6517 (0.6882) Prec 79.688% (76.412%)
Validation starts
Test: [0/79] Time 0.093 (0.093) Loss 0.7801 (0.7801) Prec 74.219%
(74.219%)
* Prec 73.380%
best acc: 73.380000
Epoch: [36][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss
0.7206 (0.7206) Prec 71.875% (71.875%)
Epoch: [36][100/391] Time 0.020 (0.026) Data 0.000 (0.006) Loss
0.6944 (0.6813) Prec 75.781% (76.686%)
Epoch: [36][200/391] Time 0.023 (0.025) Data 0.001 (0.005) Loss
0.5961 (0.6810) Prec 83.594% (76.543%)
Epoch: [36][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.5067 (0.6756) Prec 83.594% (76.742%)
Validation starts
Test: [0/79] Time 0.092 (0.092) Loss 0.8930 (0.8930) Prec 71.875%
(71.875%)
* Prec 72.890%
best acc: 73.380000
Epoch: [37][0/391] Time 0.153 (0.153) Data 0.115 (0.115) Loss
0.6339 (0.6339) Prec 78.125% (78.125%)
Epoch: [37][100/391] Time 0.021 (0.030) Data 0.000 (0.005) Loss
0.5720 (0.6422) Prec 82.812% (78.156%)
Epoch: [37][200/391] Time 0.021 (0.027) Data 0.000 (0.005) Loss
0.7490 (0.6513) Prec 75.781% (77.655%)
Epoch: [37][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.7357 (0.6537) Prec 75.000% (77.429%)
Validation starts
Test: [0/79] Time 0.094 (0.094) Loss 0.6609 (0.6609) Prec 78.125%
(78.125%)
* Prec 74.500%
best acc: 74.500000
Epoch: [38][0/391] Time 0.151 (0.151) Data 0.113 (0.113) Loss

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0.5956 (0.5956)    Prec 81.250% (81.250%)
Epoch: [38][100/391]    Time 0.022 (0.026)    Data 0.003 (0.006)    Loss
0.5639 (0.6423)    Prec 79.688% (77.924%)
Epoch: [38][200/391]    Time 0.033 (0.026)    Data 0.010 (0.006)    Loss
0.7488 (0.6545)    Prec 74.219% (77.526%)
Epoch: [38][300/391]    Time 0.031 (0.025)    Data 0.011 (0.005)    Loss
0.5919 (0.6522)    Prec 80.469% (77.660%)
Validation starts
Test: [0/79]    Time 0.093 (0.093)    Loss 0.7521 (0.7521)    Prec 73.438%
(73.438%)
* Prec 74.700%
best acc: 74.700000
Epoch: [39][0/391]    Time 0.142 (0.142)    Data 0.111 (0.111)    Loss
0.5422 (0.5422)    Prec 79.688% (79.688%)
Epoch: [39][100/391]    Time 0.023 (0.025)    Data 0.003 (0.005)    Loss
0.6404 (0.6249)    Prec 78.125% (78.651%)
Epoch: [39][200/391]    Time 0.030 (0.025)    Data 0.011 (0.005)    Loss
0.6990 (0.6370)    Prec 79.688% (78.191%)
Epoch: [39][300/391]    Time 0.026 (0.025)    Data 0.007 (0.005)    Loss
0.7943 (0.6319)    Prec 76.562% (78.330%)
Validation starts
Test: [0/79]    Time 0.091 (0.091)    Loss 0.7506 (0.7506)    Prec 77.344%
(77.344%)
* Prec 76.310%
best acc: 76.310000
Epoch: [40][0/391]    Time 0.144 (0.144)    Data 0.114 (0.114)    Loss
0.5719 (0.5719)    Prec 77.344% (77.344%)
Epoch: [40][100/391]    Time 0.031 (0.026)    Data 0.010 (0.007)    Loss
0.7160 (0.6207)    Prec 75.000% (78.179%)
Epoch: [40][200/391]    Time 0.030 (0.026)    Data 0.010 (0.006)    Loss
0.7994 (0.6197)    Prec 74.219% (78.685%)
Epoch: [40][300/391]    Time 0.027 (0.026)    Data 0.008 (0.006)    Loss
0.7015 (0.6226)    Prec 75.781% (78.667%)
Validation starts
Test: [0/79]    Time 0.091 (0.091)    Loss 0.7534 (0.7534)    Prec 75.000%
(75.000%)
* Prec 75.150%
best acc: 76.310000
Epoch: [41][0/391]    Time 0.143 (0.143)    Data 0.114 (0.114)    Loss
0.5005 (0.5005)    Prec 88.281% (88.281%)
Epoch: [41][100/391]    Time 0.027 (0.025)    Data 0.008 (0.006)    Loss
0.7610 (0.6107)    Prec 75.000% (79.154%)
Epoch: [41][200/391]    Time 0.020 (0.025)    Data 0.000 (0.005)    Loss
0.6099 (0.6103)    Prec 75.781% (79.128%)
Epoch: [41][300/391]    Time 0.020 (0.025)    Data 0.000 (0.005)    Loss
0.7287 (0.6181)    Prec 73.438% (78.725%)
Validation starts
Test: [0/79]    Time 0.092 (0.092)    Loss 0.6768 (0.6768)    Prec 77.344%

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(77.344%)

* Prec 75.310%

best acc: 76.310000

Epoch: [42] [0/391]	Time 0.147 (0.147)	Data 0.118 (0.118)	Loss
0.5077 (0.5077)	Prec 80.469% (80.469%)		
Epoch: [42] [100/391]	Time 0.032 (0.026)	Data 0.011 (0.006)	Loss
0.6793 (0.6105)	Prec 77.344% (79.100%)		
Epoch: [42] [200/391]	Time 0.031 (0.026)	Data 0.011 (0.006)	Loss
0.5456 (0.6110)	Prec 81.250% (79.089%)		
Epoch: [42] [300/391]	Time 0.030 (0.026)	Data 0.011 (0.006)	Loss
0.4736 (0.6119)	Prec 82.812% (79.091%)		

Validation starts

Test: [0/79] Time 0.092 (0.092) Loss 0.7592 (0.7592) Prec 75.000% (75.000%)

* Prec 77.060%

best acc: 77.060000

Epoch: [43] [0/391]	Time 0.144 (0.144)	Data 0.115 (0.115)	Loss
0.8023 (0.8023)	Prec 70.312% (70.312%)		
Epoch: [43] [100/391]	Time 0.033 (0.026)	Data 0.013 (0.007)	Loss
0.3910 (0.5696)	Prec 84.375% (80.314%)		
Epoch: [43] [200/391]	Time 0.030 (0.026)	Data 0.010 (0.006)	Loss
0.6746 (0.5799)	Prec 75.781% (79.859%)		
Epoch: [43] [300/391]	Time 0.035 (0.026)	Data 0.015 (0.006)	Loss
0.4523 (0.5909)	Prec 82.031% (79.685%)		

Validation starts

Test: [0/79] Time 0.095 (0.095) Loss 0.7116 (0.7116) Prec 76.562% (76.562%)

* Prec 77.280%

best acc: 77.280000

Epoch: [44] [0/391]	Time 0.142 (0.142)	Data 0.113 (0.113)	Loss
0.6965 (0.6965)	Prec 79.688% (79.688%)		
Epoch: [44] [100/391]	Time 0.020 (0.026)	Data 0.000 (0.006)	Loss
0.5574 (0.5713)	Prec 79.688% (80.569%)		
Epoch: [44] [200/391]	Time 0.020 (0.026)	Data 0.000 (0.006)	Loss
0.5435 (0.5756)	Prec 79.688% (80.239%)		
Epoch: [44] [300/391]	Time 0.030 (0.026)	Data 0.011 (0.005)	Loss
0.4042 (0.5748)	Prec 88.281% (80.246%)		

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.6483 (0.6483) Prec 73.438% (73.438%)

* Prec 77.300%

best acc: 77.300000

Epoch: [45] [0/391]	Time 0.143 (0.143)	Data 0.114 (0.114)	Loss
0.7263 (0.7263)	Prec 78.906% (78.906%)		
Epoch: [45] [100/391]	Time 0.028 (0.025)	Data 0.008 (0.005)	Loss
0.5905 (0.5825)	Prec 82.812% (80.221%)		
Epoch: [45] [200/391]	Time 0.022 (0.025)	Data 0.002 (0.005)	Loss
0.4480 (0.5806)	Prec 85.938% (80.068%)		

Epoch: [45][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.6451 (0.5848) Prec 80.469% (80.009%)
Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.7306 (0.7306) Prec 72.656%
(72.656%)
* Prec 78.070%
best acc: 78.070000
Epoch: [46][0/391] Time 0.155 (0.155) Data 0.119 (0.119) Loss
0.6209 (0.6209) Prec 78.906% (78.906%)
Epoch: [46][100/391] Time 0.031 (0.027) Data 0.012 (0.006) Loss
0.5404 (0.5600) Prec 76.562% (80.770%)
Epoch: [46][200/391] Time 0.029 (0.026) Data 0.009 (0.006) Loss
0.4604 (0.5648) Prec 84.375% (80.566%)
Epoch: [46][300/391] Time 0.030 (0.026) Data 0.010 (0.006) Loss
0.6754 (0.5676) Prec 80.469% (80.586%)
Validation starts
Test: [0/79] Time 0.088 (0.088) Loss 0.7729 (0.7729) Prec 75.000%
(75.000%)
* Prec 77.020%
best acc: 78.070000
Epoch: [47][0/391] Time 0.143 (0.143) Data 0.114 (0.114) Loss
0.6343 (0.6343) Prec 77.344% (77.344%)
Epoch: [47][100/391] Time 0.028 (0.026) Data 0.008 (0.005) Loss
0.4777 (0.5480) Prec 84.375% (81.126%)
Epoch: [47][200/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.6431 (0.5489) Prec 77.344% (81.176%)
Epoch: [47][300/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
0.5962 (0.5526) Prec 78.125% (80.944%)
Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.7938 (0.7938) Prec 74.219%
(74.219%)
* Prec 77.090%
best acc: 78.070000
Epoch: [48][0/391] Time 0.148 (0.148) Data 0.116 (0.116) Loss
0.4448 (0.4448) Prec 86.719% (86.719%)
Epoch: [48][100/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.5999 (0.5444) Prec 79.688% (81.296%)
Epoch: [48][200/391] Time 0.027 (0.025) Data 0.003 (0.005) Loss
0.6421 (0.5392) Prec 78.125% (81.573%)
Epoch: [48][300/391] Time 0.023 (0.025) Data 0.002 (0.005) Loss
0.4409 (0.5466) Prec 83.594% (81.284%)
Validation starts
Test: [0/79] Time 0.094 (0.094) Loss 0.6896 (0.6896) Prec 78.125%
(78.125%)
* Prec 77.600%
best acc: 78.070000
Epoch: [49][0/391] Time 0.143 (0.143) Data 0.114 (0.114) Loss
0.5047 (0.5047) Prec 82.812% (82.812%)

Epoch: [49][100/391] Time 0.026 (0.025) Data 0.006 (0.005) Loss
0.6626 (0.5386) Prec 74.219% (81.312%)

Epoch: [49][200/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
0.4928 (0.5362) Prec 84.375% (81.611%)

Epoch: [49][300/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.4106 (0.5352) Prec 87.500% (81.463%)

Validation starts

Test: [0/79] Time 0.099 (0.099) Loss 0.6893 (0.6893) Prec 78.125%
(78.125%)

* Prec 77.770%

best acc: 78.070000

Epoch: [50][0/391] Time 0.144 (0.144) Data 0.114 (0.114) Loss
0.4946 (0.4946) Prec 82.031% (82.031%)

Epoch: [50][100/391] Time 0.021 (0.026) Data 0.002 (0.005) Loss
0.4801 (0.5076) Prec 79.688% (82.921%)

Epoch: [50][200/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
0.5310 (0.5151) Prec 82.812% (82.408%)

Epoch: [50][300/391] Time 0.027 (0.025) Data 0.006 (0.005) Loss
0.4676 (0.5196) Prec 84.375% (82.249%)

Validation starts

Test: [0/79] Time 0.097 (0.097) Loss 0.6568 (0.6568) Prec 77.344%
(77.344%)

* Prec 76.620%

best acc: 78.070000

Epoch: [51][0/391] Time 0.155 (0.155) Data 0.118 (0.118) Loss
0.4674 (0.4674) Prec 84.375% (84.375%)

Epoch: [51][100/391] Time 0.022 (0.027) Data 0.002 (0.006) Loss
0.4061 (0.5097) Prec 85.156% (82.519%)

Epoch: [51][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.5719 (0.5120) Prec 78.906% (82.490%)

Epoch: [51][300/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss
0.4040 (0.5134) Prec 86.719% (82.465%)

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.6755 (0.6755) Prec 76.562%
(76.562%)

* Prec 78.340%

best acc: 78.340000

Epoch: [52][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss
0.3852 (0.3852) Prec 87.500% (87.500%)

Epoch: [52][100/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss
0.4998 (0.4990) Prec 80.469% (83.130%)

Epoch: [52][200/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss
0.6483 (0.5040) Prec 76.562% (82.797%)

Epoch: [52][300/391] Time 0.028 (0.026) Data 0.007 (0.004) Loss
0.5015 (0.5050) Prec 82.812% (82.691%)

Validation starts

Test: [0/79] Time 0.101 (0.101) Loss 0.6042 (0.6042) Prec 77.344%
(77.344%)

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* Prec 77.510%
best acc: 78.340000
Epoch: [53][0/391]      Time 0.145 (0.145)      Data 0.116 (0.116)      Loss
0.5397 (0.5397)      Prec 81.250% (81.250%)
Epoch: [53][100/391]    Time 0.021 (0.026)      Data 0.001 (0.005)      Loss
0.5137 (0.4923)      Prec 82.812% (82.921%)
Epoch: [53][200/391]    Time 0.021 (0.025)      Data 0.001 (0.005)      Loss
0.4760 (0.4952)      Prec 85.938% (82.976%)
Epoch: [53][300/391]    Time 0.027 (0.026)      Data 0.006 (0.004)      Loss
0.5368 (0.4946)      Prec 86.719% (82.937%)
Validation starts
Test: [0/79]      Time 0.101 (0.101)      Loss 0.6763 (0.6763)      Prec 76.562%
(76.562%)
* Prec 78.650%
best acc: 78.650000
Epoch: [54][0/391]      Time 0.155 (0.155)      Data 0.124 (0.124)      Loss
0.3632 (0.3632)      Prec 90.625% (90.625%)
Epoch: [54][100/391]    Time 0.027 (0.026)      Data 0.005 (0.005)      Loss
0.6519 (0.4879)      Prec 80.469% (83.222%)
Epoch: [54][200/391]    Time 0.026 (0.025)      Data 0.006 (0.004)      Loss
0.5068 (0.4865)      Prec 86.719% (83.590%)
Epoch: [54][300/391]    Time 0.022 (0.025)      Data 0.002 (0.004)      Loss
0.5944 (0.4907)      Prec 80.469% (83.425%)
Validation starts
Test: [0/79]      Time 0.090 (0.090)      Loss 0.5542 (0.5542)      Prec 81.250%
(81.250%)
* Prec 79.660%
best acc: 79.660000
Epoch: [55][0/391]      Time 0.144 (0.144)      Data 0.115 (0.115)      Loss
0.3864 (0.3864)      Prec 85.938% (85.938%)
Epoch: [55][100/391]    Time 0.027 (0.025)      Data 0.007 (0.005)      Loss
0.4460 (0.4643)      Prec 87.500% (84.004%)
Epoch: [55][200/391]    Time 0.022 (0.025)      Data 0.002 (0.005)      Loss
0.4648 (0.4845)      Prec 86.719% (83.415%)
Epoch: [55][300/391]    Time 0.029 (0.025)      Data 0.008 (0.005)      Loss
0.5622 (0.4849)      Prec 81.250% (83.300%)
Validation starts
Test: [0/79]      Time 0.095 (0.095)      Loss 0.5654 (0.5654)      Prec 82.031%
(82.031%)
* Prec 79.970%
best acc: 79.970000
Epoch: [56][0/391]      Time 0.148 (0.148)      Data 0.118 (0.118)      Loss
0.4278 (0.4278)      Prec 87.500% (87.500%)
Epoch: [56][100/391]    Time 0.029 (0.026)      Data 0.010 (0.006)      Loss
0.4019 (0.4526)      Prec 87.500% (84.267%)
Epoch: [56][200/391]    Time 0.028 (0.025)      Data 0.008 (0.005)      Loss
0.4371 (0.4701)      Prec 86.719% (83.780%)
Epoch: [56][300/391]    Time 0.025 (0.025)      Data 0.002 (0.005)      Loss

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0.4680 (0.4735) Prec 82.031% (83.812%)

Validation starts

Test: [0/79] Time 0.094 (0.094) Loss 0.4046 (0.4046) Prec 88.281% (88.281%)

* Prec 78.930%

best acc: 79.970000

Epoch: [57][0/391]	Time 0.144 (0.144)	Data 0.115 (0.115)	Loss
0.4388 (0.4388)	Prec 83.594% (83.594%)		
Epoch: [57][100/391]	Time 0.028 (0.026)	Data 0.008 (0.006)	Loss
0.5890 (0.4601)	Prec 81.250% (84.460%)		
Epoch: [57][200/391]	Time 0.028 (0.025)	Data 0.009 (0.005)	Loss
0.6368 (0.4632)	Prec 79.688% (84.309%)		
Epoch: [57][300/391]	Time 0.021 (0.025)	Data 0.000 (0.005)	Loss
0.4785 (0.4639)	Prec 82.031% (84.188%)		

Validation starts

Test: [0/79] Time 0.092 (0.092) Loss 0.5671 (0.5671) Prec 80.469% (80.469%)

* Prec 80.580%

best acc: 80.580000

Epoch: [58][0/391]	Time 0.140 (0.140)	Data 0.111 (0.111)	Loss
0.5143 (0.5143)	Prec 85.156% (85.156%)		
Epoch: [58][100/391]	Time 0.029 (0.025)	Data 0.009 (0.006)	Loss
0.4475 (0.4449)	Prec 83.594% (84.940%)		
Epoch: [58][200/391]	Time 0.022 (0.025)	Data 0.002 (0.005)	Loss
0.3949 (0.4590)	Prec 86.719% (84.530%)		
Epoch: [58][300/391]	Time 0.028 (0.025)	Data 0.004 (0.005)	Loss
0.4711 (0.4595)	Prec 83.594% (84.481%)		

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.7280 (0.7280) Prec 78.906% (78.906%)

* Prec 77.760%

best acc: 80.580000

Epoch: [59][0/391]	Time 0.143 (0.143)	Data 0.114 (0.114)	Loss
0.5735 (0.5735)	Prec 80.469% (80.469%)		
Epoch: [59][100/391]	Time 0.022 (0.026)	Data 0.002 (0.005)	Loss
0.3067 (0.4357)	Prec 88.281% (84.901%)		
Epoch: [59][200/391]	Time 0.028 (0.025)	Data 0.009 (0.005)	Loss
0.4268 (0.4395)	Prec 85.156% (84.919%)		
Epoch: [59][300/391]	Time 0.026 (0.025)	Data 0.007 (0.005)	Loss
0.4388 (0.4414)	Prec 85.938% (84.902%)		

Validation starts

Test: [0/79] Time 0.098 (0.098) Loss 0.6339 (0.6339) Prec 76.562% (76.562%)

* Prec 80.020%

best acc: 80.580000

Epoch: [60][0/391]	Time 0.143 (0.143)	Data 0.113 (0.113)	Loss
0.3683 (0.3683)	Prec 87.500% (87.500%)		
Epoch: [60][100/391]	Time 0.029 (0.026)	Data 0.008 (0.006)	Loss

0.3217 (0.4262) Prec 89.844% (85.381%)
Epoch: [60][200/391] Time 0.030 (0.025) Data 0.010 (0.005) Loss
0.4081 (0.4308) Prec 82.031% (85.316%)
Epoch: [60][300/391] Time 0.025 (0.025) Data 0.003 (0.005) Loss
0.4466 (0.4439) Prec 83.594% (84.972%)
Validation starts
Test: [0/79] Time 0.098 (0.098) Loss 0.5645 (0.5645) Prec 82.812%
(82.812%)
* Prec 78.690%
best acc: 80.580000
Epoch: [61][0/391] Time 0.152 (0.152) Data 0.114 (0.114) Loss
0.3872 (0.3872) Prec 86.719% (86.719%)
Epoch: [61][100/391] Time 0.026 (0.026) Data 0.006 (0.005) Loss
0.4693 (0.4259) Prec 82.031% (85.257%)
Epoch: [61][200/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.3134 (0.4335) Prec 88.281% (85.160%)
Epoch: [61][300/391] Time 0.037 (0.025) Data 0.012 (0.005) Loss
0.3942 (0.4368) Prec 85.938% (85.021%)
Validation starts
Test: [0/79] Time 0.094 (0.094) Loss 0.5436 (0.5436) Prec 82.812%
(82.812%)
* Prec 80.620%
best acc: 80.620000
Epoch: [62][0/391] Time 0.151 (0.151) Data 0.113 (0.113) Loss
0.4152 (0.4152) Prec 85.938% (85.938%)
Epoch: [62][100/391] Time 0.027 (0.026) Data 0.007 (0.005) Loss
0.3468 (0.4244) Prec 87.500% (85.504%)
Epoch: [62][200/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.4213 (0.4194) Prec 86.719% (85.588%)
Epoch: [62][300/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.3505 (0.4229) Prec 89.062% (85.475%)
Validation starts
Test: [0/79] Time 0.093 (0.093) Loss 0.4989 (0.4989) Prec 83.594%
(83.594%)
* Prec 81.390%
best acc: 81.390000
Epoch: [63][0/391] Time 0.152 (0.152) Data 0.115 (0.115) Loss
0.5083 (0.5083) Prec 82.031% (82.031%)
Epoch: [63][100/391] Time 0.030 (0.027) Data 0.010 (0.006) Loss
0.4512 (0.4255) Prec 84.375% (85.412%)
Epoch: [63][200/391] Time 0.027 (0.026) Data 0.006 (0.005) Loss
0.6996 (0.4188) Prec 76.562% (85.627%)
Epoch: [63][300/391] Time 0.028 (0.025) Data 0.004 (0.005) Loss
0.4395 (0.4137) Prec 85.938% (85.699%)
Validation starts
Test: [0/79] Time 0.096 (0.096) Loss 0.4458 (0.4458) Prec 83.594%
(83.594%)
* Prec 81.730%

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best acc: 81.730000
Epoch: [64][0/391]      Time 0.152 (0.152)      Data 0.122 (0.122)      Loss
0.2968 (0.2968)      Prec 87.500% (87.500%)
Epoch: [64][100/391]    Time 0.030 (0.026)      Data 0.010 (0.006)      Loss
0.4203 (0.3930)      Prec 85.938% (86.479%)
Epoch: [64][200/391]    Time 0.032 (0.026)      Data 0.012 (0.006)      Loss
0.4902 (0.3977)      Prec 85.938% (86.194%)
Epoch: [64][300/391]    Time 0.035 (0.026)      Data 0.011 (0.006)      Loss
0.2812 (0.4084)      Prec 90.625% (85.956%)
Validation starts
Test: [0/79]      Time 0.091 (0.091)      Loss 0.4899 (0.4899)      Prec 82.812%
(82.812%)
* Prec 81.470%
best acc: 81.730000
Epoch: [65][0/391]      Time 0.144 (0.144)      Data 0.114 (0.114)      Loss
0.3826 (0.3826)      Prec 84.375% (84.375%)
Epoch: [65][100/391]    Time 0.027 (0.026)      Data 0.007 (0.005)      Loss
0.4009 (0.3915)      Prec 84.375% (86.541%)
Epoch: [65][200/391]    Time 0.022 (0.025)      Data 0.002 (0.005)      Loss
0.4568 (0.3973)      Prec 82.812% (86.427%)
Epoch: [65][300/391]    Time 0.020 (0.025)      Data 0.000 (0.005)      Loss
0.3071 (0.3991)      Prec 90.625% (86.433%)
Validation starts
Test: [0/79]      Time 0.095 (0.095)      Loss 0.4494 (0.4494)      Prec 85.156%
(85.156%)
* Prec 82.200%
best acc: 82.200000
Epoch: [66][0/391]      Time 0.149 (0.149)      Data 0.120 (0.120)      Loss
0.4149 (0.4149)      Prec 85.156% (85.156%)
Epoch: [66][100/391]    Time 0.030 (0.026)      Data 0.008 (0.005)      Loss
0.3276 (0.3991)      Prec 90.625% (86.494%)
Epoch: [66][200/391]    Time 0.029 (0.025)      Data 0.009 (0.005)      Loss
0.2975 (0.3959)      Prec 90.625% (86.629%)
Epoch: [66][300/391]    Time 0.029 (0.025)      Data 0.010 (0.005)      Loss
0.3739 (0.3973)      Prec 86.719% (86.573%)
Validation starts
Test: [0/79]      Time 0.095 (0.095)      Loss 0.6366 (0.6366)      Prec 78.125%
(78.125%)
* Prec 80.140%
best acc: 82.200000
Epoch: [67][0/391]      Time 0.148 (0.148)      Data 0.119 (0.119)      Loss
0.3112 (0.3112)      Prec 91.406% (91.406%)
Epoch: [67][100/391]    Time 0.031 (0.026)      Data 0.011 (0.006)      Loss
0.3355 (0.3811)      Prec 85.156% (86.897%)
Epoch: [67][200/391]    Time 0.030 (0.026)      Data 0.006 (0.006)      Loss
0.4178 (0.3844)      Prec 86.719% (86.812%)
Epoch: [67][300/391]    Time 0.032 (0.026)      Data 0.011 (0.006)      Loss
0.3813 (0.3871)      Prec 83.594% (86.729%)

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Validation starts

Test: [0/79] Time 0.099 (0.099) Loss 0.4612 (0.4612) Prec 82.812%
(82.812%)

* Prec 81.890%

best acc: 82.200000

Epoch: [68][0/391] Time 0.153 (0.153) Data 0.115 (0.115) Loss
0.2525 (0.2525) Prec 93.750% (93.750%)

Epoch: [68][100/391] Time 0.028 (0.026) Data 0.007 (0.005) Loss
0.4779 (0.3801) Prec 85.938% (86.997%)

Epoch: [68][200/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
0.4061 (0.3759) Prec 89.062% (87.232%)

Epoch: [68][300/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.4524 (0.3822) Prec 86.719% (87.098%)

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.4119 (0.4119) Prec 85.938%
(85.938%)

* Prec 82.210%

best acc: 82.210000

Epoch: [69][0/391] Time 0.148 (0.148) Data 0.119 (0.119) Loss
0.3457 (0.3457) Prec 88.281% (88.281%)

Epoch: [69][100/391] Time 0.028 (0.027) Data 0.008 (0.005) Loss
0.3503 (0.3702) Prec 91.406% (87.554%)

Epoch: [69][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.3980 (0.3771) Prec 85.156% (87.185%)

Epoch: [69][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.3088 (0.3762) Prec 91.406% (87.277%)

Validation starts

Test: [0/79] Time 0.095 (0.095) Loss 0.4071 (0.4071) Prec 88.281%
(88.281%)

* Prec 82.100%

best acc: 82.210000

Epoch: [70][0/391] Time 0.145 (0.145) Data 0.116 (0.116) Loss
0.3637 (0.3637) Prec 86.719% (86.719%)

Epoch: [70][100/391] Time 0.024 (0.026) Data 0.004 (0.005) Loss
0.3319 (0.3585) Prec 88.281% (87.717%)

Epoch: [70][200/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.3038 (0.3614) Prec 91.406% (87.648%)

Epoch: [70][300/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
0.3845 (0.3701) Prec 89.062% (87.388%)

Validation starts

Test: [0/79] Time 0.094 (0.094) Loss 0.4700 (0.4700) Prec 82.031%
(82.031%)

* Prec 82.940%

best acc: 82.940000

Epoch: [71][0/391] Time 0.155 (0.155) Data 0.116 (0.116) Loss
0.2193 (0.2193) Prec 92.969% (92.969%)

Epoch: [71][100/391] Time 0.030 (0.027) Data 0.011 (0.006) Loss
0.3900 (0.3522) Prec 89.844% (87.825%)

Epoch: [71][200/391] Time 0.028 (0.026) Data 0.008 (0.006) Loss
0.3371 (0.3573) Prec 89.844% (87.679%)

Epoch: [71][300/391] Time 0.028 (0.026) Data 0.008 (0.005) Loss
0.2855 (0.3589) Prec 86.719% (87.666%)

Validation starts
Test: [0/79] Time 0.093 (0.093) Loss 0.4170 (0.4170) Prec 86.719%
(86.719%)
* Prec 83.260%
best acc: 83.260000

Epoch: [72][0/391] Time 0.144 (0.144) Data 0.114 (0.114) Loss
0.2639 (0.2639) Prec 89.062% (89.062%)

Epoch: [72][100/391] Time 0.029 (0.026) Data 0.009 (0.006) Loss
0.2515 (0.3465) Prec 93.750% (88.119%)

Epoch: [72][200/391] Time 0.028 (0.025) Data 0.007 (0.005) Loss
0.3648 (0.3552) Prec 88.281% (87.858%)

Epoch: [72][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.2977 (0.3597) Prec 89.844% (87.747%)

Validation starts
Test: [0/79] Time 0.095 (0.095) Loss 0.4340 (0.4340) Prec 84.375%
(84.375%)
* Prec 83.750%
best acc: 83.750000

Epoch: [73][0/391] Time 0.150 (0.150) Data 0.113 (0.113) Loss
0.3016 (0.3016) Prec 89.062% (89.062%)

Epoch: [73][100/391] Time 0.022 (0.028) Data 0.002 (0.007) Loss
0.3157 (0.3522) Prec 88.281% (87.918%)

Epoch: [73][200/391] Time 0.022 (0.027) Data 0.002 (0.006) Loss
0.3430 (0.3593) Prec 88.281% (87.702%)

Epoch: [73][300/391] Time 0.021 (0.026) Data 0.002 (0.005) Loss
0.2482 (0.3611) Prec 92.188% (87.684%)

Validation starts
Test: [0/79] Time 0.102 (0.102) Loss 0.3698 (0.3698) Prec 84.375%
(84.375%)
* Prec 83.390%
best acc: 83.750000

Epoch: [74][0/391] Time 0.144 (0.144) Data 0.115 (0.115) Loss
0.2537 (0.2537) Prec 91.406% (91.406%)

Epoch: [74][100/391] Time 0.028 (0.026) Data 0.008 (0.005) Loss
0.4458 (0.3509) Prec 82.812% (88.096%)

Epoch: [74][200/391] Time 0.031 (0.025) Data 0.010 (0.005) Loss
0.2387 (0.3539) Prec 92.188% (87.947%)

Epoch: [74][300/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.3626 (0.3540) Prec 88.281% (87.970%)

Validation starts
Test: [0/79] Time 0.095 (0.095) Loss 0.4008 (0.4008) Prec 85.156%
(85.156%)
* Prec 84.170%
best acc: 84.170000

Epoch: [75][0/391] Time 0.141 (0.141) Data 0.112 (0.112) Loss
0.1950 (0.1950) Prec 92.969% (92.969%)

Epoch: [75][100/391] Time 0.034 (0.026) Data 0.013 (0.006) Loss
0.4370 (0.3405) Prec 85.938% (88.390%)

Epoch: [75][200/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.2597 (0.3494) Prec 91.406% (88.083%)

Epoch: [75][300/391] Time 0.030 (0.025) Data 0.010 (0.005) Loss
0.4799 (0.3537) Prec 83.594% (87.827%)

Validation starts

Test: [0/79] Time 0.092 (0.092) Loss 0.4240 (0.4240) Prec 85.938%
(85.938%)

* Prec 83.810%

best acc: 84.170000

Epoch: [76][0/391] Time 0.153 (0.153) Data 0.116 (0.116) Loss
0.2965 (0.2965) Prec 87.500% (87.500%)

Epoch: [76][100/391] Time 0.029 (0.026) Data 0.010 (0.006) Loss
0.3011 (0.3252) Prec 89.062% (88.745%)

Epoch: [76][200/391] Time 0.028 (0.026) Data 0.008 (0.005) Loss
0.3661 (0.3304) Prec 88.281% (88.787%)

Epoch: [76][300/391] Time 0.029 (0.025) Data 0.010 (0.005) Loss
0.3357 (0.3342) Prec 91.406% (88.536%)

Validation starts

Test: [0/79] Time 0.098 (0.098) Loss 0.3675 (0.3675) Prec 87.500%
(87.500%)

* Prec 83.000%

best acc: 84.170000

Epoch: [77][0/391] Time 0.151 (0.151) Data 0.114 (0.114) Loss
0.4329 (0.4329) Prec 88.281% (88.281%)

Epoch: [77][100/391] Time 0.026 (0.026) Data 0.007 (0.005) Loss
0.2596 (0.3255) Prec 90.625% (88.614%)

Epoch: [77][200/391] Time 0.025 (0.025) Data 0.002 (0.005) Loss
0.4407 (0.3337) Prec 85.156% (88.305%)

Epoch: [77][300/391] Time 0.022 (0.025) Data 0.002 (0.004) Loss
0.3470 (0.3385) Prec 88.281% (88.227%)

Validation starts

Test: [0/79] Time 0.098 (0.098) Loss 0.3894 (0.3894) Prec 85.156%
(85.156%)

* Prec 85.140%

best acc: 85.140000

Epoch: [78][0/391] Time 0.143 (0.143) Data 0.113 (0.113) Loss
0.2514 (0.2514) Prec 91.406% (91.406%)

Epoch: [78][100/391] Time 0.028 (0.026) Data 0.007 (0.005) Loss
0.2885 (0.3104) Prec 91.406% (89.604%)

Epoch: [78][200/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.2846 (0.3235) Prec 90.625% (88.954%)

Epoch: [78][300/391] Time 0.029 (0.025) Data 0.008 (0.005) Loss
0.1313 (0.3252) Prec 96.094% (88.889%)

Validation starts

Test: [0/79] Time 0.099 (0.099) Loss 0.5222 (0.5222) Prec 84.375%
(84.375%)

* Prec 83.260%

best acc: 85.140000

Epoch: [79][0/391] Time 0.145 (0.145) Data 0.116 (0.116) Loss
0.5374 (0.5374) Prec 85.156% (85.156%)

Epoch: [79][100/391] Time 0.027 (0.025) Data 0.007 (0.005) Loss
0.3202 (0.3224) Prec 87.500% (89.163%)

Epoch: [79][200/391] Time 0.025 (0.025) Data 0.006 (0.005) Loss
0.2847 (0.3257) Prec 92.188% (88.965%)

Epoch: [79][300/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.3403 (0.3237) Prec 89.062% (88.974%)

Validation starts

Test: [0/79] Time 0.095 (0.095) Loss 0.4564 (0.4564) Prec 87.500%
(87.500%)

* Prec 83.930%

best acc: 85.140000

Epoch: [80][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss
0.2818 (0.2818) Prec 85.156% (85.156%)

Epoch: [80][100/391] Time 0.028 (0.027) Data 0.009 (0.006) Loss
0.2503 (0.3122) Prec 89.844% (89.109%)

Epoch: [80][200/391] Time 0.028 (0.026) Data 0.007 (0.005) Loss
0.2764 (0.3198) Prec 90.625% (88.911%)

Epoch: [80][300/391] Time 0.029 (0.025) Data 0.010 (0.005) Loss
0.3183 (0.3199) Prec 89.062% (88.943%)

Validation starts

Test: [0/79] Time 0.097 (0.097) Loss 0.4343 (0.4343) Prec 85.938%
(85.938%)

* Prec 84.760%

best acc: 85.140000

Epoch: [81][0/391] Time 0.156 (0.156) Data 0.119 (0.119) Loss
0.3483 (0.3483) Prec 86.719% (86.719%)

Epoch: [81][100/391] Time 0.021 (0.027) Data 0.000 (0.006) Loss
0.3957 (0.3042) Prec 89.062% (89.372%)

Epoch: [81][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.3123 (0.3061) Prec 90.625% (89.381%)

Epoch: [81][300/391] Time 0.021 (0.025) Data 0.001 (0.005) Loss
0.2698 (0.3135) Prec 91.406% (89.120%)

Validation starts

Test: [0/79] Time 0.097 (0.097) Loss 0.3517 (0.3517) Prec 86.719%
(86.719%)

* Prec 84.640%

best acc: 85.140000

Epoch: [82][0/391] Time 0.153 (0.153) Data 0.115 (0.115) Loss
0.2693 (0.2693) Prec 90.625% (90.625%)

Epoch: [82][100/391] Time 0.028 (0.026) Data 0.007 (0.005) Loss
0.4217 (0.2934) Prec 88.281% (90.060%)

Epoch: [82][200/391] Time 0.027 (0.025) Data 0.005 (0.004) Loss

0.2684 (0.3043) Prec 89.844% (89.603%)
Epoch: [82][300/391] Time 0.029 (0.025) Data 0.007 (0.004) Loss
0.2445 (0.3074) Prec 92.188% (89.491%)
Validation starts
Test: [0/79] Time 0.094 (0.094) Loss 0.3663 (0.3663) Prec 86.719%
(86.719%)
* Prec 85.100%
best acc: 85.140000
Epoch: [83][0/391] Time 0.141 (0.141) Data 0.112 (0.112) Loss
0.3437 (0.3437) Prec 87.500% (87.500%)
Epoch: [83][100/391] Time 0.028 (0.026) Data 0.008 (0.005) Loss
0.2954 (0.3048) Prec 89.062% (89.380%)
Epoch: [83][200/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
0.2910 (0.3041) Prec 87.500% (89.541%)
Epoch: [83][300/391] Time 0.029 (0.025) Data 0.010 (0.005) Loss
0.2365 (0.3014) Prec 89.062% (89.621%)
Validation starts
Test: [0/79] Time 0.104 (0.104) Loss 0.4972 (0.4972) Prec 85.156%
(85.156%)
* Prec 84.370%
best acc: 85.140000
Epoch: [84][0/391] Time 0.150 (0.150) Data 0.120 (0.120) Loss
0.2763 (0.2763) Prec 89.844% (89.844%)
Epoch: [84][100/391] Time 0.030 (0.026) Data 0.011 (0.006) Loss
0.3467 (0.2881) Prec 90.625% (90.192%)
Epoch: [84][200/391] Time 0.028 (0.026) Data 0.007 (0.005) Loss
0.2315 (0.2914) Prec 92.188% (90.050%)
Epoch: [84][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.4365 (0.2985) Prec 84.375% (89.709%)
Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.3745 (0.3745) Prec 88.281%
(88.281%)
* Prec 85.590%
best acc: 85.590000
Epoch: [85][0/391] Time 0.150 (0.150) Data 0.112 (0.112) Loss
0.2088 (0.2088) Prec 92.969% (92.969%)
Epoch: [85][100/391] Time 0.031 (0.027) Data 0.011 (0.005) Loss
0.2774 (0.2820) Prec 92.188% (90.401%)
Epoch: [85][200/391] Time 0.028 (0.027) Data 0.007 (0.004) Loss
0.3912 (0.2910) Prec 88.281% (89.984%)
Epoch: [85][300/391] Time 0.028 (0.026) Data 0.008 (0.004) Loss
0.2564 (0.2925) Prec 89.844% (90.049%)
Validation starts
Test: [0/79] Time 0.092 (0.092) Loss 0.5776 (0.5776) Prec 82.031%
(82.031%)
* Prec 83.500%
best acc: 85.590000
Epoch: [86][0/391] Time 0.144 (0.144) Data 0.115 (0.115) Loss

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0.2377 (0.2377)    Prec 92.188% (92.188%)
Epoch: [86][100/391]    Time 0.028 (0.027)    Data 0.009 (0.006)    Loss
0.1610 (0.2793)    Prec 95.312% (90.246%)
Epoch: [86][200/391]    Time 0.028 (0.026)    Data 0.008 (0.005)    Loss
0.2108 (0.2876)    Prec 92.188% (90.116%)
Epoch: [86][300/391]    Time 0.029 (0.025)    Data 0.009 (0.005)    Loss
0.2204 (0.2918)    Prec 93.750% (89.984%)
Validation starts
Test: [0/79]    Time 0.091 (0.091)    Loss 0.3958 (0.3958)    Prec 86.719%
(86.719%)
* Prec 84.860%
best acc: 85.590000
Epoch: [87][0/391]    Time 0.143 (0.143)    Data 0.113 (0.113)    Loss
0.2165 (0.2165)    Prec 91.406% (91.406%)
Epoch: [87][100/391]    Time 0.028 (0.025)    Data 0.008 (0.005)    Loss
0.2062 (0.2741)    Prec 92.188% (90.586%)
Epoch: [87][200/391]    Time 0.029 (0.025)    Data 0.008 (0.004)    Loss
0.3551 (0.2821)    Prec 89.062% (90.295%)
Epoch: [87][300/391]    Time 0.020 (0.025)    Data 0.000 (0.004)    Loss
0.2412 (0.2849)    Prec 91.406% (90.249%)
Validation starts
Test: [0/79]    Time 0.092 (0.092)    Loss 0.5113 (0.5113)    Prec 84.375%
(84.375%)
* Prec 85.100%
best acc: 85.590000
Epoch: [88][0/391]    Time 0.152 (0.152)    Data 0.122 (0.122)    Loss
0.2840 (0.2840)    Prec 90.625% (90.625%)
Epoch: [88][100/391]    Time 0.028 (0.026)    Data 0.007 (0.005)    Loss
0.3263 (0.2771)    Prec 88.281% (90.710%)
Epoch: [88][200/391]    Time 0.020 (0.026)    Data 0.000 (0.005)    Loss
0.3310 (0.2810)    Prec 88.281% (90.501%)
Epoch: [88][300/391]    Time 0.021 (0.025)    Data 0.000 (0.005)    Loss
0.3581 (0.2849)    Prec 89.062% (90.358%)
Validation starts
Test: [0/79]    Time 0.096 (0.096)    Loss 0.5277 (0.5277)    Prec 85.156%
(85.156%)
* Prec 84.000%
best acc: 85.590000
Epoch: [89][0/391]    Time 0.152 (0.152)    Data 0.122 (0.122)    Loss
0.1987 (0.1987)    Prec 92.969% (92.969%)
Epoch: [89][100/391]    Time 0.031 (0.027)    Data 0.011 (0.006)    Loss
0.2764 (0.2894)    Prec 89.844% (90.169%)
Epoch: [89][200/391]    Time 0.021 (0.027)    Data 0.002 (0.005)    Loss
0.2495 (0.2792)    Prec 92.188% (90.481%)
Epoch: [89][300/391]    Time 0.031 (0.026)    Data 0.012 (0.006)    Loss
0.2288 (0.2804)    Prec 92.969% (90.441%)
Validation starts
Test: [0/79]    Time 0.089 (0.089)    Loss 0.5899 (0.5899)    Prec 82.812%

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(82.812%)
* Prec 84.630%
best acc: 85.590000
Epoch: [90][0/391]      Time 0.167 (0.167)      Data 0.137 (0.137)      Loss
0.3426 (0.3426)      Prec 89.844% (89.844%)
Epoch: [90][100/391]    Time 0.028 (0.026)      Data 0.008 (0.006)      Loss
0.2053 (0.2806)      Prec 91.406% (90.470%)
Epoch: [90][200/391]    Time 0.021 (0.026)      Data 0.000 (0.006)      Loss
0.2466 (0.2777)      Prec 92.969% (90.536%)
Epoch: [90][300/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
0.3080 (0.2801)      Prec 89.844% (90.537%)
Validation starts
Test: [0/79]      Time 0.094 (0.094)      Loss 0.5310 (0.5310)      Prec 82.812%
(82.812%)
* Prec 82.980%
best acc: 85.590000
Epoch: [91][0/391]      Time 0.146 (0.146)      Data 0.116 (0.116)      Loss
0.1918 (0.1918)      Prec 92.188% (92.188%)
Epoch: [91][100/391]    Time 0.022 (0.026)      Data 0.002 (0.006)      Loss
0.2909 (0.2647)      Prec 89.844% (90.873%)
Epoch: [91][200/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
0.2755 (0.2653)      Prec 89.844% (90.808%)
Epoch: [91][300/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
0.2255 (0.2692)      Prec 91.406% (90.685%)
Validation starts
Test: [0/79]      Time 0.094 (0.094)      Loss 0.4010 (0.4010)      Prec 84.375%
(84.375%)
* Prec 85.350%
best acc: 85.590000
Epoch: [92][0/391]      Time 0.160 (0.160)      Data 0.120 (0.120)      Loss
0.2168 (0.2168)      Prec 91.406% (91.406%)
Epoch: [92][100/391]    Time 0.029 (0.026)      Data 0.009 (0.005)      Loss
0.2196 (0.2615)      Prec 93.750% (91.105%)
Epoch: [92][200/391]    Time 0.029 (0.026)      Data 0.010 (0.005)      Loss
0.2294 (0.2688)      Prec 92.188% (90.668%)
Epoch: [92][300/391]    Time 0.030 (0.026)      Data 0.011 (0.005)      Loss
0.1780 (0.2739)      Prec 94.531% (90.477%)
Validation starts
Test: [0/79]      Time 0.095 (0.095)      Loss 0.4138 (0.4138)      Prec 87.500%
(87.500%)
* Prec 84.820%
best acc: 85.590000
Epoch: [93][0/391]      Time 0.151 (0.151)      Data 0.119 (0.119)      Loss
0.2589 (0.2589)      Prec 89.844% (89.844%)
Epoch: [93][100/391]    Time 0.030 (0.026)      Data 0.009 (0.006)      Loss
0.3066 (0.2488)      Prec 90.625% (91.306%)
Epoch: [93][200/391]    Time 0.027 (0.026)      Data 0.007 (0.005)      Loss
0.3036 (0.2569)      Prec 88.281% (91.115%)

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Epoch: [93][300/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
0.2405 (0.2659) Prec 89.844% (90.843%)
Validation starts
Test: [0/79] Time 0.090 (0.090) Loss 0.4092 (0.4092) Prec 85.156%
(85.156%)
* Prec 87.150%
best acc: 87.150000
Epoch: [94][0/391] Time 0.151 (0.151) Data 0.120 (0.120) Loss
0.2121 (0.2121) Prec 94.531% (94.531%)
Epoch: [94][100/391] Time 0.028 (0.026) Data 0.007 (0.005) Loss
0.2434 (0.2448) Prec 92.969% (91.576%)
Epoch: [94][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.4086 (0.2530) Prec 85.938% (91.402%)
Epoch: [94][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.3486 (0.2576) Prec 89.844% (91.248%)
Validation starts
Test: [0/79] Time 0.094 (0.094) Loss 0.4689 (0.4689) Prec 88.281%
(88.281%)
* Prec 84.560%
best acc: 87.150000
Epoch: [95][0/391] Time 0.153 (0.153) Data 0.115 (0.115) Loss
0.2616 (0.2616) Prec 92.188% (92.188%)
Epoch: [95][100/391] Time 0.029 (0.026) Data 0.009 (0.005) Loss
0.2430 (0.2661) Prec 92.969% (91.035%)
Epoch: [95][200/391] Time 0.028 (0.026) Data 0.008 (0.005) Loss
0.2515 (0.2621) Prec 89.844% (91.045%)
Epoch: [95][300/391] Time 0.026 (0.025) Data 0.005 (0.005) Loss
0.2709 (0.2639) Prec 89.844% (90.999%)
Validation starts
Test: [0/79] Time 0.094 (0.094) Loss 0.2535 (0.2535) Prec 91.406%
(91.406%)
* Prec 85.490%
best acc: 87.150000
Epoch: [96][0/391] Time 0.154 (0.154) Data 0.122 (0.122) Loss
0.2277 (0.2277) Prec 90.625% (90.625%)
Epoch: [96][100/391] Time 0.029 (0.027) Data 0.008 (0.006) Loss
0.2530 (0.2312) Prec 90.625% (91.948%)
Epoch: [96][200/391] Time 0.030 (0.026) Data 0.010 (0.006) Loss
0.1734 (0.2468) Prec 93.750% (91.422%)
Epoch: [96][300/391] Time 0.034 (0.026) Data 0.014 (0.006) Loss
0.1479 (0.2457) Prec 95.312% (91.536%)
Validation starts
Test: [0/79] Time 0.095 (0.095) Loss 0.4253 (0.4253) Prec 88.281%
(88.281%)
* Prec 85.210%
best acc: 87.150000
Epoch: [97][0/391] Time 0.154 (0.154) Data 0.125 (0.125) Loss
0.2200 (0.2200) Prec 92.188% (92.188%)

Epoch: [97][100/391] Time 0.028 (0.028) Data 0.008 (0.005) Loss
 0.1816 (0.2556) Prec 92.969% (91.136%)
 Epoch: [97][200/391] Time 0.030 (0.026) Data 0.010 (0.005) Loss
 0.2916 (0.2601) Prec 89.062% (91.103%)
 Epoch: [97][300/391] Time 0.030 (0.026) Data 0.010 (0.005) Loss
 0.2540 (0.2587) Prec 91.406% (91.131%)
 Validation starts
 Test: [0/79] Time 0.088 (0.088) Loss 0.3552 (0.3552) Prec 89.844%
 (89.844%)
 * Prec 84.770%
 best acc: 87.150000
 Epoch: [98][0/391] Time 0.151 (0.151) Data 0.114 (0.114) Loss
 0.2440 (0.2440) Prec 95.312% (95.312%)
 Epoch: [98][100/391] Time 0.030 (0.027) Data 0.009 (0.006) Loss
 0.3154 (0.2432) Prec 89.844% (91.368%)
 Epoch: [98][200/391] Time 0.026 (0.025) Data 0.007 (0.005) Loss
 0.1786 (0.2553) Prec 94.531% (91.006%)
 Epoch: [98][300/391] Time 0.029 (0.025) Data 0.008 (0.005) Loss
 0.1884 (0.2545) Prec 92.969% (91.110%)
 Validation starts
 Test: [0/79] Time 0.090 (0.090) Loss 0.4180 (0.4180) Prec 83.594%
 (83.594%)
 * Prec 84.160%
 best acc: 87.150000
 Epoch: [99][0/391] Time 0.148 (0.148) Data 0.118 (0.118) Loss
 0.3591 (0.3591) Prec 86.719% (86.719%)
 Epoch: [99][100/391] Time 0.022 (0.026) Data 0.002 (0.005) Loss
 0.2861 (0.2445) Prec 89.844% (91.399%)
 Epoch: [99][200/391] Time 0.027 (0.026) Data 0.007 (0.005) Loss
 0.3990 (0.2499) Prec 86.719% (91.418%)
 Epoch: [99][300/391] Time 0.024 (0.025) Data 0.004 (0.005) Loss
 0.2583 (0.2510) Prec 89.844% (91.305%)
 Validation starts
 Test: [0/79] Time 0.094 (0.094) Loss 0.3992 (0.3992) Prec 88.281%
 (88.281%)
 * Prec 84.340%
 best acc: 87.150000
 Epoch: [100][0/391] Time 0.148 (0.148) Data 0.118 (0.118) Loss
 0.2767 (0.2767) Prec 88.281% (88.281%)
 Epoch: [100][100/391] Time 0.031 (0.027) Data 0.008 (0.005) Loss
 0.2184 (0.2035) Prec 90.625% (92.915%)
 Epoch: [100][200/391] Time 0.031 (0.026) Data 0.010 (0.005) Loss
 0.1638 (0.1924) Prec 96.094% (93.493%)
 Epoch: [100][300/391] Time 0.030 (0.026) Data 0.010 (0.005) Loss
 0.1385 (0.1859) Prec 96.094% (93.654%)
 Validation starts
 Test: [0/79] Time 0.092 (0.092) Loss 0.3068 (0.3068) Prec 92.188%
 (92.188%)

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* Prec 88.430%
best acc: 88.430000
Epoch: [101][0/391]      Time 0.149 (0.149)      Data 0.113 (0.113)      Loss
0.0888 (0.0888)      Prec 96.875% (96.875%)
Epoch: [101][100/391]    Time 0.021 (0.027)      Data 0.000 (0.005)      Loss
0.1277 (0.1621)      Prec 96.094% (94.369%)
Epoch: [101][200/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
0.1720 (0.1605)      Prec 93.750% (94.516%)
Epoch: [101][300/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
0.1575 (0.1591)      Prec 94.531% (94.534%)
Validation starts
Test: [0/79]      Time 0.098 (0.098)      Loss 0.2421 (0.2421)      Prec 93.750%
(93.750%)
* Prec 88.680%
best acc: 88.680000
Epoch: [102][0/391]      Time 0.148 (0.148)      Data 0.119 (0.119)      Loss
0.1440 (0.1440)      Prec 95.312% (95.312%)
Epoch: [102][100/391]    Time 0.028 (0.026)      Data 0.007 (0.006)      Loss
0.1747 (0.1402)      Prec 95.312% (95.088%)
Epoch: [102][200/391]    Time 0.022 (0.026)      Data 0.002 (0.005)      Loss
0.0603 (0.1438)      Prec 97.656% (95.017%)
Epoch: [102][300/391]    Time 0.020 (0.025)      Data 0.000 (0.005)      Loss
0.0660 (0.1456)      Prec 98.438% (94.931%)
Validation starts
Test: [0/79]      Time 0.094 (0.094)      Loss 0.2353 (0.2353)      Prec 93.750%
(93.750%)
* Prec 88.230%
best acc: 88.680000
Epoch: [103][0/391]      Time 0.149 (0.149)      Data 0.119 (0.119)      Loss
0.1344 (0.1344)      Prec 96.875% (96.875%)
Epoch: [103][100/391]    Time 0.029 (0.027)      Data 0.009 (0.006)      Loss
0.0986 (0.1290)      Prec 96.875% (95.599%)
Epoch: [103][200/391]    Time 0.030 (0.026)      Data 0.009 (0.006)      Loss
0.1461 (0.1343)      Prec 96.094% (95.414%)
Epoch: [103][300/391]    Time 0.039 (0.026)      Data 0.019 (0.006)      Loss
0.0685 (0.1369)      Prec 99.219% (95.338%)
Validation starts
Test: [0/79]      Time 0.095 (0.095)      Loss 0.4084 (0.4084)      Prec 88.281%
(88.281%)
* Prec 88.620%
best acc: 88.680000
Epoch: [104][0/391]      Time 0.150 (0.150)      Data 0.120 (0.120)      Loss
0.1377 (0.1377)      Prec 96.094% (96.094%)
Epoch: [104][100/391]    Time 0.022 (0.026)      Data 0.002 (0.006)      Loss
0.2081 (0.1346)      Prec 92.188% (95.359%)
Epoch: [104][200/391]    Time 0.032 (0.025)      Data 0.008 (0.005)      Loss
0.1702 (0.1347)      Prec 92.969% (95.382%)
Epoch: [104][300/391]    Time 0.029 (0.025)      Data 0.010 (0.005)      Loss

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0.1628 (0.1362) Prec 92.969% (95.323%)
Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.3132 (0.3132) Prec 88.281%
(88.281%)
* Prec 88.470%
best acc: 88.680000
Epoch: [105][0/391] Time 0.145 (0.145) Data 0.116 (0.116) Loss
0.0712 (0.0712) Prec 97.656% (97.656%)
Epoch: [105][100/391] Time 0.020 (0.026) Data 0.001 (0.005) Loss
0.1333 (0.1342) Prec 94.531% (95.413%)
Epoch: [105][200/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
0.1009 (0.1333) Prec 96.875% (95.402%)
Epoch: [105][300/391] Time 0.022 (0.025) Data 0.002 (0.005) Loss
0.1935 (0.1339) Prec 90.625% (95.362%)
Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.2604 (0.2604) Prec 88.281%
(88.281%)
* Prec 88.650%
best acc: 88.680000
Epoch: [106][0/391] Time 0.147 (0.147) Data 0.117 (0.117) Loss
0.0878 (0.0878) Prec 96.875% (96.875%)
Epoch: [106][100/391] Time 0.020 (0.027) Data 0.000 (0.006) Loss
0.1758 (0.1271) Prec 95.312% (95.777%)
Epoch: [106][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.2205 (0.1269) Prec 92.188% (95.756%)
Epoch: [106][300/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss
0.1209 (0.1275) Prec 95.312% (95.681%)
Validation starts
Test: [0/79] Time 0.090 (0.090) Loss 0.2733 (0.2733) Prec 92.969%
(92.969%)
* Prec 89.100%
best acc: 89.100000
Epoch: [107][0/391] Time 0.149 (0.149) Data 0.120 (0.120) Loss
0.1537 (0.1537) Prec 95.312% (95.312%)
Epoch: [107][100/391] Time 0.028 (0.027) Data 0.008 (0.006) Loss
0.0744 (0.1204) Prec 96.875% (95.769%)
Epoch: [107][200/391] Time 0.022 (0.026) Data 0.002 (0.006) Loss
0.0754 (0.1201) Prec 98.438% (95.783%)
Epoch: [107][300/391] Time 0.020 (0.026) Data 0.000 (0.006) Loss
0.1103 (0.1209) Prec 96.094% (95.826%)
Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.3098 (0.3098) Prec 89.844%
(89.844%)
* Prec 88.660%
best acc: 89.100000
Epoch: [108][0/391] Time 0.155 (0.155) Data 0.118 (0.118) Loss
0.0790 (0.0790) Prec 97.656% (97.656%)
Epoch: [108][100/391] Time 0.022 (0.027) Data 0.000 (0.005) Loss

0.1742 (0.1221) Prec 93.750% (95.769%)
Epoch: [108][200/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss
0.1296 (0.1198) Prec 95.312% (95.779%)
Epoch: [108][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.0761 (0.1203) Prec 97.656% (95.821%)
Validation starts
Test: [0/79] Time 0.090 (0.090) Loss 0.2891 (0.2891) Prec 90.625%
(90.625%)
* Prec 88.900%
best acc: 89.100000
Epoch: [109][0/391] Time 0.152 (0.152) Data 0.114 (0.114) Loss
0.0806 (0.0806) Prec 97.656% (97.656%)
Epoch: [109][100/391] Time 0.020 (0.027) Data 0.000 (0.005) Loss
0.0621 (0.1148) Prec 97.656% (96.233%)
Epoch: [109][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.1293 (0.1160) Prec 94.531% (96.098%)
Epoch: [109][300/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss
0.0660 (0.1157) Prec 99.219% (96.094%)
Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.3526 (0.3526) Prec 90.625%
(90.625%)
* Prec 88.620%
best acc: 89.100000
Epoch: [110][0/391] Time 0.149 (0.149) Data 0.119 (0.119) Loss
0.0642 (0.0642) Prec 97.656% (97.656%)
Epoch: [110][100/391] Time 0.020 (0.027) Data 0.001 (0.006) Loss
0.1891 (0.1145) Prec 96.094% (96.086%)
Epoch: [110][200/391] Time 0.021 (0.026) Data 0.000 (0.006) Loss
0.1933 (0.1187) Prec 94.531% (95.896%)
Epoch: [110][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.1122 (0.1178) Prec 97.656% (95.987%)
Validation starts
Test: [0/79] Time 0.089 (0.089) Loss 0.2512 (0.2512) Prec 90.625%
(90.625%)
* Prec 88.920%
best acc: 89.100000
Epoch: [111][0/391] Time 0.157 (0.157) Data 0.120 (0.120) Loss
0.1129 (0.1129) Prec 93.750% (93.750%)
Epoch: [111][100/391] Time 0.031 (0.027) Data 0.010 (0.006) Loss
0.1012 (0.1120) Prec 96.875% (96.187%)
Epoch: [111][200/391] Time 0.030 (0.026) Data 0.010 (0.006) Loss
0.1119 (0.1141) Prec 96.094% (96.144%)
Epoch: [111][300/391] Time 0.030 (0.026) Data 0.010 (0.006) Loss
0.1103 (0.1115) Prec 96.094% (96.195%)
Validation starts
Test: [0/79] Time 0.091 (0.091) Loss 0.3000 (0.3000) Prec 92.188%
(92.188%)
* Prec 89.060%

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best acc: 89.100000
Epoch: [112][0/391]      Time 0.152 (0.152)      Data 0.114 (0.114)      Loss
0.1846 (0.1846)      Prec 95.312% (95.312%)
Epoch: [112][100/391]    Time 0.020 (0.027)      Data 0.000 (0.006)      Loss
0.0688 (0.1092)      Prec 96.875% (96.140%)
Epoch: [112][200/391]    Time 0.020 (0.026)      Data 0.000 (0.006)      Loss
0.1026 (0.1150)      Prec 96.875% (95.985%)
Epoch: [112][300/391]    Time 0.020 (0.026)      Data 0.000 (0.006)      Loss
0.1222 (0.1144)      Prec 95.312% (96.052%)
Validation starts
Test: [0/79]      Time 0.093 (0.093)      Loss 0.2395 (0.2395)      Prec 94.531%
(94.531%)
* Prec 88.920%
best acc: 89.100000
Epoch: [113][0/391]      Time 0.146 (0.146)      Data 0.115 (0.115)      Loss
0.1877 (0.1877)      Prec 93.750% (93.750%)
Epoch: [113][100/391]    Time 0.020 (0.027)      Data 0.000 (0.005)      Loss
0.1327 (0.1110)      Prec 95.312% (96.024%)
Epoch: [113][200/391]    Time 0.023 (0.026)      Data 0.000 (0.005)      Loss
0.0751 (0.1140)      Prec 97.656% (95.989%)
Epoch: [113][300/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
0.1364 (0.1133)      Prec 95.312% (96.003%)
Validation starts
Test: [0/79]      Time 0.094 (0.094)      Loss 0.3414 (0.3414)      Prec 91.406%
(91.406%)
* Prec 88.710%
best acc: 89.100000
Epoch: [114][0/391]      Time 0.148 (0.148)      Data 0.119 (0.119)      Loss
0.1369 (0.1369)      Prec 93.750% (93.750%)
Epoch: [114][100/391]    Time 0.025 (0.027)      Data 0.005 (0.005)      Loss
0.1485 (0.1136)      Prec 93.750% (96.001%)
Epoch: [114][200/391]    Time 0.020 (0.026)      Data 0.000 (0.005)      Loss
0.1218 (0.1113)      Prec 95.312% (96.117%)
Epoch: [114][300/391]    Time 0.028 (0.025)      Data 0.008 (0.005)      Loss
0.1923 (0.1094)      Prec 95.312% (96.182%)
Validation starts
Test: [0/79]      Time 0.090 (0.090)      Loss 0.2795 (0.2795)      Prec 90.625%
(90.625%)
* Prec 88.810%
best acc: 89.100000
Epoch: [115][0/391]      Time 0.143 (0.143)      Data 0.114 (0.114)      Loss
0.1082 (0.1082)      Prec 96.875% (96.875%)
Epoch: [115][100/391]    Time 0.034 (0.027)      Data 0.010 (0.006)      Loss
0.1789 (0.0985)      Prec 94.531% (96.535%)
Epoch: [115][200/391]    Time 0.030 (0.026)      Data 0.010 (0.006)      Loss
0.0900 (0.1021)      Prec 97.656% (96.541%)
Epoch: [115][300/391]    Time 0.030 (0.026)      Data 0.010 (0.005)      Loss
0.1744 (0.1060)      Prec 94.531% (96.346%)

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Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.2470 (0.2470) Prec 93.750%
(93.750%)

* Prec 88.740%

best acc: 89.100000

Epoch: [116][0/391]	Time 0.151 (0.151)	Data 0.122 (0.122)	Loss
0.1615 (0.1615)	Prec 94.531% (94.531%)		
Epoch: [116][100/391]	Time 0.020 (0.027)	Data 0.001 (0.005)	Loss
0.0984 (0.1132)	Prec 94.531% (96.132%)		
Epoch: [116][200/391]	Time 0.020 (0.025)	Data 0.000 (0.005)	Loss
0.1069 (0.1086)	Prec 95.312% (96.210%)		
Epoch: [116][300/391]	Time 0.028 (0.025)	Data 0.001 (0.005)	Loss
0.1201 (0.1102)	Prec 95.312% (96.182%)		

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.2462 (0.2462) Prec 92.188%
(92.188%)

* Prec 88.880%

best acc: 89.100000

Epoch: [117][0/391]	Time 0.160 (0.160)	Data 0.119 (0.119)	Loss
0.0830 (0.0830)	Prec 96.875% (96.875%)		
Epoch: [117][100/391]	Time 0.029 (0.027)	Data 0.009 (0.006)	Loss
0.1715 (0.1074)	Prec 95.312% (96.287%)		
Epoch: [117][200/391]	Time 0.030 (0.026)	Data 0.009 (0.005)	Loss
0.1418 (0.1109)	Prec 93.750% (96.125%)		
Epoch: [117][300/391]	Time 0.028 (0.025)	Data 0.007 (0.005)	Loss
0.0561 (0.1118)	Prec 98.438% (96.117%)		

Validation starts

Test: [0/79] Time 0.091 (0.091) Loss 0.3477 (0.3477) Prec 90.625%
(90.625%)

* Prec 88.730%

best acc: 89.100000

Epoch: [118][0/391]	Time 0.144 (0.144)	Data 0.115 (0.115)	Loss
0.0923 (0.0923)	Prec 96.875% (96.875%)		
Epoch: [118][100/391]	Time 0.026 (0.026)	Data 0.006 (0.005)	Loss
0.0574 (0.1112)	Prec 99.219% (96.202%)		
Epoch: [118][200/391]	Time 0.020 (0.025)	Data 0.000 (0.005)	Loss
0.1581 (0.1079)	Prec 94.531% (96.276%)		
Epoch: [118][300/391]	Time 0.021 (0.025)	Data 0.000 (0.005)	Loss
0.1190 (0.1067)	Prec 96.094% (96.330%)		

Validation starts

Test: [0/79] Time 0.102 (0.102) Loss 0.2048 (0.2048) Prec 92.969%
(92.969%)

* Prec 89.030%

best acc: 89.100000

Epoch: [119][0/391]	Time 0.144 (0.144)	Data 0.115 (0.115)	Loss
0.0847 (0.0847)	Prec 96.094% (96.094%)		
Epoch: [119][100/391]	Time 0.029 (0.026)	Data 0.009 (0.005)	Loss
0.0686 (0.1098)	Prec 97.656% (96.063%)		

Epoch: [119][200/391] Time 0.028 (0.025) Data 0.008 (0.005) Loss
 0.0620 (0.1089) Prec 97.656% (96.074%)
 Epoch: [119][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
 0.1068 (0.1085) Prec 97.656% (96.073%)
 Validation starts
 Test: [0/79] Time 0.094 (0.094) Loss 0.2406 (0.2406) Prec 92.969%
 (92.969%)
 * Prec 88.920%
 best acc: 89.100000
 Epoch: [120][0/391] Time 0.147 (0.147) Data 0.117 (0.117) Loss
 0.0692 (0.0692) Prec 98.438% (98.438%)
 Epoch: [120][100/391] Time 0.020 (0.027) Data 0.000 (0.005) Loss
 0.1361 (0.0995) Prec 96.875% (96.620%)
 Epoch: [120][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
 0.0642 (0.1012) Prec 97.656% (96.521%)
 Epoch: [120][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
 0.0975 (0.0992) Prec 95.312% (96.543%)
 Validation starts
 Test: [0/79] Time 0.094 (0.094) Loss 0.2685 (0.2685) Prec 90.625%
 (90.625%)
 * Prec 89.130%
 best acc: 89.130000
 Epoch: [121][0/391] Time 0.144 (0.144) Data 0.115 (0.115) Loss
 0.0709 (0.0709) Prec 95.312% (95.312%)
 Epoch: [121][100/391] Time 0.034 (0.027) Data 0.013 (0.005) Loss
 0.0998 (0.0910) Prec 94.531% (96.821%)
 Epoch: [121][200/391] Time 0.031 (0.026) Data 0.011 (0.005) Loss
 0.0650 (0.0987) Prec 98.438% (96.587%)
 Epoch: [121][300/391] Time 0.030 (0.026) Data 0.010 (0.005) Loss
 0.0777 (0.0987) Prec 97.656% (96.641%)
 Validation starts
 Test: [0/79] Time 0.102 (0.102) Loss 0.3280 (0.3280) Prec 90.625%
 (90.625%)
 * Prec 89.240%
 best acc: 89.240000
 Epoch: [122][0/391] Time 0.151 (0.151) Data 0.122 (0.122) Loss
 0.0811 (0.0811) Prec 98.438% (98.438%)
 Epoch: [122][100/391] Time 0.028 (0.026) Data 0.008 (0.006) Loss
 0.1452 (0.0985) Prec 94.531% (96.697%)
 Epoch: [122][200/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
 0.0882 (0.0994) Prec 96.094% (96.591%)
 Epoch: [122][300/391] Time 0.020 (0.025) Data 0.001 (0.005) Loss
 0.0617 (0.0972) Prec 96.094% (96.649%)
 Validation starts
 Test: [0/79] Time 0.093 (0.093) Loss 0.2413 (0.2413) Prec 91.406%
 (91.406%)
 * Prec 88.950%
 best acc: 89.240000

Epoch: [123][0/391] Time 0.152 (0.152) Data 0.115 (0.115) Loss
0.0524 (0.0524) Prec 98.438% (98.438%)

Epoch: [123][100/391] Time 0.028 (0.027) Data 0.008 (0.006) Loss
0.0298 (0.0868) Prec 98.438% (97.006%)

Epoch: [123][200/391] Time 0.029 (0.026) Data 0.009 (0.005) Loss
0.0771 (0.0933) Prec 97.656% (96.937%)

Epoch: [123][300/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.0708 (0.0952) Prec 96.875% (96.810%)

Validation starts

Test: [0/79] Time 0.093 (0.093) Loss 0.2918 (0.2918) Prec 92.188%
(92.188%)

* Prec 89.110%

best acc: 89.240000

Epoch: [124][0/391] Time 0.153 (0.153) Data 0.116 (0.116) Loss
0.0617 (0.0617) Prec 97.656% (97.656%)

Epoch: [124][100/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.0796 (0.0944) Prec 98.438% (97.006%)

Epoch: [124][200/391] Time 0.023 (0.026) Data 0.000 (0.005) Loss
0.0857 (0.0958) Prec 97.656% (96.852%)

Epoch: [124][300/391] Time 0.020 (0.025) Data 0.000 (0.005) Loss
0.0770 (0.0936) Prec 96.875% (96.917%)

Validation starts

Test: [0/79] Time 0.095 (0.095) Loss 0.2835 (0.2835) Prec 89.844%
(89.844%)

* Prec 89.000%

best acc: 89.240000

Epoch: [125][0/391] Time 0.151 (0.151) Data 0.114 (0.114) Loss
0.0582 (0.0582) Prec 96.875% (96.875%)

Epoch: [125][100/391] Time 0.033 (0.029) Data 0.011 (0.005) Loss
0.0589 (0.0919) Prec 96.875% (96.790%)

Epoch: [125][200/391] Time 0.022 (0.027) Data 0.002 (0.005) Loss
0.0671 (0.0897) Prec 96.875% (96.852%)

Epoch: [125][300/391] Time 0.021 (0.026) Data 0.001 (0.005) Loss
0.0407 (0.0909) Prec 100.000% (96.854%)

Validation starts

Test: [0/79] Time 0.099 (0.099) Loss 0.2573 (0.2573) Prec 91.406%
(91.406%)

* Prec 89.150%

best acc: 89.240000

Epoch: [126][0/391] Time 0.144 (0.144) Data 0.115 (0.115) Loss
0.0775 (0.0775) Prec 96.875% (96.875%)

Epoch: [126][100/391] Time 0.022 (0.026) Data 0.002 (0.006) Loss
0.0970 (0.0916) Prec 96.094% (97.037%)

Epoch: [126][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.0869 (0.0950) Prec 95.312% (96.883%)

Epoch: [126][300/391] Time 0.021 (0.025) Data 0.002 (0.005) Loss
0.0979 (0.0927) Prec 96.094% (96.901%)

Validation starts

Test: [0/79] Time 0.097 (0.097) Loss 0.2932 (0.2932) Prec 93.750%
(93.750%)

* Prec 89.230%

best acc: 89.240000

Epoch: [127][0/391] Time 0.151 (0.151) Data 0.113 (0.113) Loss
0.0752 (0.0752) Prec 96.875% (96.875%)

Epoch: [127][100/391] Time 0.028 (0.028) Data 0.009 (0.005) Loss
0.1675 (0.0953) Prec 94.531% (96.481%)

Epoch: [127][200/391] Time 0.022 (0.026) Data 0.002 (0.005) Loss
0.1224 (0.0934) Prec 97.656% (96.669%)

Epoch: [127][300/391] Time 0.022 (0.026) Data 0.002 (0.005) Loss
0.1083 (0.0929) Prec 96.094% (96.691%)

Validation starts

Test: [0/79] Time 0.098 (0.098) Loss 0.2029 (0.2029) Prec 91.406%
(91.406%)

* Prec 89.130%

best acc: 89.240000

Epoch: [128][0/391] Time 0.157 (0.157) Data 0.120 (0.120) Loss
0.0899 (0.0899) Prec 96.875% (96.875%)

Epoch: [128][100/391] Time 0.029 (0.027) Data 0.009 (0.005) Loss
0.0569 (0.0942) Prec 98.438% (96.999%)

Epoch: [128][200/391] Time 0.028 (0.026) Data 0.008 (0.005) Loss
0.0647 (0.0948) Prec 96.875% (96.793%)

Epoch: [128][300/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.1325 (0.0945) Prec 96.094% (96.792%)

Validation starts

Test: [0/79] Time 0.091 (0.091) Loss 0.2183 (0.2183) Prec 93.750%
(93.750%)

* Prec 89.100%

best acc: 89.240000

Epoch: [129][0/391] Time 0.147 (0.147) Data 0.118 (0.118) Loss
0.0926 (0.0926) Prec 96.875% (96.875%)

Epoch: [129][100/391] Time 0.023 (0.028) Data 0.000 (0.005) Loss
0.0825 (0.1000) Prec 97.656% (96.511%)

Epoch: [129][200/391] Time 0.021 (0.027) Data 0.000 (0.005) Loss
0.0868 (0.0948) Prec 95.312% (96.603%)

Epoch: [129][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.1580 (0.0955) Prec 95.312% (96.641%)

Validation starts

Test: [0/79] Time 0.090 (0.090) Loss 0.2307 (0.2307) Prec 90.625%
(90.625%)

* Prec 89.220%

best acc: 89.240000

Epoch: [130][0/391] Time 0.142 (0.142) Data 0.112 (0.112) Loss
0.2230 (0.2230) Prec 92.969% (92.969%)

Epoch: [130][100/391] Time 0.033 (0.027) Data 0.010 (0.006) Loss
0.1060 (0.0972) Prec 96.094% (96.682%)

Epoch: [130][200/391] Time 0.026 (0.026) Data 0.006 (0.005) Loss

0.0845 (0.0935) Prec 96.875% (96.809%)
Epoch: [130][300/391] Time 0.029 (0.025) Data 0.009 (0.005) Loss
0.0658 (0.0923) Prec 96.875% (96.836%)
Validation starts
Test: [0/79] Time 0.094 (0.094) Loss 0.3180 (0.3180) Prec 90.625%
(90.625%)
* Prec 89.270%
best acc: 89.270000
Epoch: [131][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss
0.0684 (0.0684) Prec 97.656% (97.656%)
Epoch: [131][100/391] Time 0.020 (0.028) Data 0.000 (0.005) Loss
0.0880 (0.0984) Prec 96.094% (96.805%)
Epoch: [131][200/391] Time 0.031 (0.027) Data 0.012 (0.005) Loss
0.1137 (0.0968) Prec 96.094% (96.665%)
Epoch: [131][300/391] Time 0.028 (0.026) Data 0.008 (0.005) Loss
0.1356 (0.0934) Prec 96.875% (96.740%)
Validation starts
Test: [0/79] Time 0.093 (0.093) Loss 0.2601 (0.2601) Prec 92.969%
(92.969%)
* Prec 89.250%
best acc: 89.270000
Epoch: [132][0/391] Time 0.143 (0.143) Data 0.114 (0.114) Loss
0.0839 (0.0839) Prec 97.656% (97.656%)
Epoch: [132][100/391] Time 0.033 (0.027) Data 0.011 (0.005) Loss
0.0687 (0.0897) Prec 97.656% (96.976%)
Epoch: [132][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.0514 (0.0883) Prec 97.656% (97.019%)
Epoch: [132][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.0713 (0.0891) Prec 96.875% (96.968%)
Validation starts
Test: [0/79] Time 0.093 (0.093) Loss 0.2734 (0.2734) Prec 92.188%
(92.188%)
* Prec 89.260%
best acc: 89.270000
Epoch: [133][0/391] Time 0.147 (0.147) Data 0.117 (0.117) Loss
0.0291 (0.0291) Prec 100.000% (100.000%)
Epoch: [133][100/391] Time 0.020 (0.027) Data 0.000 (0.005) Loss
0.1243 (0.0925) Prec 96.094% (96.860%)
Epoch: [133][200/391] Time 0.029 (0.026) Data 0.009 (0.005) Loss
0.0789 (0.0908) Prec 96.875% (96.867%)
Epoch: [133][300/391] Time 0.027 (0.025) Data 0.006 (0.005) Loss
0.0476 (0.0907) Prec 99.219% (96.867%)
Validation starts
Test: [0/79] Time 0.099 (0.099) Loss 0.2973 (0.2973) Prec 90.625%
(90.625%)
* Prec 89.100%
best acc: 89.270000
Epoch: [134][0/391] Time 0.148 (0.148) Data 0.117 (0.117) Loss

```

0.0782 (0.0782)    Prec 96.875% (96.875%)
Epoch: [134][100/391]    Time 0.035 (0.026)    Data 0.013 (0.005)    Loss
0.1188 (0.0941)    Prec 95.312% (96.720%)
Epoch: [134][200/391]    Time 0.020 (0.026)    Data 0.000 (0.005)    Loss
0.0773 (0.0941)    Prec 96.875% (96.723%)
Epoch: [134][300/391]    Time 0.020 (0.025)    Data 0.000 (0.005)    Loss
0.1211 (0.0943)    Prec 93.750% (96.748%)
Validation starts
Test: [0/79]    Time 0.094 (0.094)    Loss 0.2822 (0.2822)    Prec 91.406%
(91.406%)
* Prec 88.690%
best acc: 89.270000
Epoch: [135][0/391]    Time 0.151 (0.151)    Data 0.114 (0.114)    Loss
0.0889 (0.0889)    Prec 98.438% (98.438%)
Epoch: [135][100/391]    Time 0.026 (0.026)    Data 0.006 (0.005)    Loss
0.0480 (0.0918)    Prec 98.438% (96.821%)
Epoch: [135][200/391]    Time 0.029 (0.026)    Data 0.010 (0.005)    Loss
0.0932 (0.0871)    Prec 96.875% (97.003%)
Epoch: [135][300/391]    Time 0.029 (0.026)    Data 0.009 (0.005)    Loss
0.0976 (0.0904)    Prec 96.875% (96.922%)
Validation starts
Test: [0/79]    Time 0.097 (0.097)    Loss 0.2642 (0.2642)    Prec 89.844%
(89.844%)
* Prec 89.420%
best acc: 89.420000
Epoch: [136][0/391]    Time 0.141 (0.141)    Data 0.112 (0.112)    Loss
0.1274 (0.1274)    Prec 95.312% (95.312%)
Epoch: [136][100/391]    Time 0.029 (0.026)    Data 0.001 (0.005)    Loss
0.0588 (0.0870)    Prec 96.094% (96.937%)
Epoch: [136][200/391]    Time 0.021 (0.026)    Data 0.001 (0.004)    Loss
0.1077 (0.0884)    Prec 96.875% (96.926%)
Epoch: [136][300/391]    Time 0.021 (0.026)    Data 0.000 (0.004)    Loss
0.0482 (0.0893)    Prec 98.438% (96.942%)
Validation starts
Test: [0/79]    Time 0.099 (0.099)    Loss 0.2467 (0.2467)    Prec 92.188%
(92.188%)
* Prec 89.040%
best acc: 89.420000
Epoch: [137][0/391]    Time 0.144 (0.144)    Data 0.114 (0.114)    Loss
0.0990 (0.0990)    Prec 96.094% (96.094%)
Epoch: [137][100/391]    Time 0.028 (0.026)    Data 0.006 (0.005)    Loss
0.1081 (0.0909)    Prec 96.094% (96.829%)
Epoch: [137][200/391]    Time 0.032 (0.026)    Data 0.004 (0.004)    Loss
0.0604 (0.0948)    Prec 96.094% (96.657%)
Epoch: [137][300/391]    Time 0.024 (0.026)    Data 0.004 (0.004)    Loss
0.0835 (0.0929)    Prec 96.094% (96.761%)
Validation starts
Test: [0/79]    Time 0.102 (0.102)    Loss 0.2652 (0.2652)    Prec 91.406%

```

```

(91.406%)
* Prec 88.850%
best acc: 89.420000
Epoch: [138][0/391]      Time 0.155 (0.155)      Data 0.125 (0.125)      Loss
0.0513 (0.0513)      Prec 96.875% (96.875%)
Epoch: [138][100/391]    Time 0.029 (0.027)      Data 0.000 (0.005)      Loss
0.0270 (0.0896)      Prec 100.000% (96.790%)
Epoch: [138][200/391]    Time 0.023 (0.028)      Data 0.001 (0.004)      Loss
0.1070 (0.0868)      Prec 97.656% (96.914%)
Epoch: [138][300/391]    Time 0.022 (0.027)      Data 0.001 (0.004)      Loss
0.0978 (0.0877)      Prec 96.094% (96.922%)
Validation starts
Test: [0/79]      Time 0.098 (0.098)      Loss 0.2213 (0.2213)      Prec 94.531%
(94.531%)
* Prec 89.180%
best acc: 89.420000
Epoch: [139][0/391]      Time 0.154 (0.154)      Data 0.124 (0.124)      Loss
0.1099 (0.1099)      Prec 94.531% (94.531%)
Epoch: [139][100/391]    Time 0.022 (0.026)      Data 0.000 (0.005)      Loss
0.1052 (0.0876)      Prec 95.312% (96.666%)
Epoch: [139][200/391]    Time 0.030 (0.026)      Data 0.009 (0.005)      Loss
0.0940 (0.0869)      Prec 94.531% (96.793%)
Epoch: [139][300/391]    Time 0.030 (0.026)      Data 0.009 (0.004)      Loss
0.0543 (0.0882)      Prec 98.438% (96.815%)
Validation starts
Test: [0/79]      Time 0.103 (0.103)      Loss 0.2610 (0.2610)      Prec 92.188%
(92.188%)
* Prec 89.160%
best acc: 89.420000
Epoch: [140][0/391]      Time 0.146 (0.146)      Data 0.116 (0.116)      Loss
0.0639 (0.0639)      Prec 98.438% (98.438%)
Epoch: [140][100/391]    Time 0.026 (0.027)      Data 0.002 (0.005)      Loss
0.0463 (0.0950)      Prec 99.219% (96.689%)
Epoch: [140][200/391]    Time 0.025 (0.026)      Data 0.000 (0.004)      Loss
0.0623 (0.0898)      Prec 97.656% (96.887%)
Epoch: [140][300/391]    Time 0.021 (0.026)      Data 0.000 (0.004)      Loss
0.1070 (0.0918)      Prec 97.656% (96.782%)
Validation starts
Test: [0/79]      Time 0.099 (0.099)      Loss 0.2958 (0.2958)      Prec 91.406%
(91.406%)
* Prec 89.130%
best acc: 89.420000
Epoch: [141][0/391]      Time 0.145 (0.145)      Data 0.116 (0.116)      Loss
0.0447 (0.0447)      Prec 97.656% (97.656%)
Epoch: [141][100/391]    Time 0.029 (0.028)      Data 0.008 (0.005)      Loss
0.0743 (0.0894)      Prec 97.656% (96.798%)
Epoch: [141][200/391]    Time 0.027 (0.026)      Data 0.007 (0.005)      Loss
0.1329 (0.0870)      Prec 96.875% (96.992%)

```

Epoch: [141][300/391] Time 0.030 (0.026) Data 0.009 (0.004) Loss
0.0581 (0.0878) Prec 97.656% (96.945%)
Validation starts
Test: [0/79] Time 0.097 (0.097) Loss 0.2527 (0.2527) Prec 93.750%
(93.750%)
* Prec 89.080%
best acc: 89.420000
Epoch: [142][0/391] Time 0.146 (0.146) Data 0.117 (0.117) Loss
0.0591 (0.0591) Prec 96.875% (96.875%)
Epoch: [142][100/391] Time 0.031 (0.027) Data 0.002 (0.005) Loss
0.0911 (0.0856) Prec 96.094% (97.115%)
Epoch: [142][200/391] Time 0.021 (0.026) Data 0.000 (0.005) Loss
0.0726 (0.0903) Prec 96.094% (96.828%)
Epoch: [142][300/391] Time 0.020 (0.026) Data 0.000 (0.004) Loss
0.0690 (0.0916) Prec 98.438% (96.800%)
Validation starts
Test: [0/79] Time 0.098 (0.098) Loss 0.2903 (0.2903) Prec 92.188%
(92.188%)
* Prec 89.510%
best acc: 89.510000
Epoch: [143][0/391] Time 0.151 (0.151) Data 0.114 (0.114) Loss
0.0960 (0.0960) Prec 95.312% (95.312%)
Epoch: [143][100/391] Time 0.021 (0.027) Data 0.000 (0.004) Loss
0.0979 (0.0916) Prec 96.094% (96.829%)
Epoch: [143][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.0490 (0.0874) Prec 98.438% (96.964%)
Epoch: [143][300/391] Time 0.026 (0.026) Data 0.001 (0.005) Loss
0.0581 (0.0895) Prec 96.875% (96.911%)
Validation starts
Test: [0/79] Time 0.100 (0.100) Loss 0.2956 (0.2956) Prec 92.188%
(92.188%)
* Prec 89.330%
best acc: 89.510000
Epoch: [144][0/391] Time 0.142 (0.142) Data 0.113 (0.113) Loss
0.1245 (0.1245) Prec 97.656% (97.656%)
Epoch: [144][100/391] Time 0.021 (0.027) Data 0.000 (0.005) Loss
0.0759 (0.0886) Prec 97.656% (96.836%)
Epoch: [144][200/391] Time 0.030 (0.026) Data 0.008 (0.005) Loss
0.0559 (0.0872) Prec 96.875% (96.879%)
Epoch: [144][300/391] Time 0.027 (0.026) Data 0.007 (0.005) Loss
0.0915 (0.0883) Prec 96.094% (96.867%)
Validation starts
Test: [0/79] Time 0.096 (0.096) Loss 0.2805 (0.2805) Prec 90.625%
(90.625%)
* Prec 89.200%
best acc: 89.510000
Epoch: [145][0/391] Time 0.152 (0.152) Data 0.123 (0.123) Loss
0.1310 (0.1310) Prec 95.312% (95.312%)

Epoch: [145][100/391] Time 0.030 (0.027) Data 0.010 (0.005) Loss
0.0338 (0.0877) Prec 99.219% (96.999%)

Epoch: [145][200/391] Time 0.037 (0.028) Data 0.007 (0.005) Loss
0.0673 (0.0884) Prec 99.219% (96.980%)

Epoch: [145][300/391] Time 0.031 (0.027) Data 0.010 (0.005) Loss
0.1844 (0.0880) Prec 95.312% (97.018%)

Validation starts

Test: [0/79] Time 0.100 (0.100) Loss 0.2549 (0.2549) Prec 92.188%
(92.188%)

* Prec 88.900%

best acc: 89.510000

Epoch: [146][0/391] Time 0.154 (0.154) Data 0.115 (0.115) Loss
0.0783 (0.0783) Prec 97.656% (97.656%)

Epoch: [146][100/391] Time 0.021 (0.027) Data 0.000 (0.005) Loss
0.0492 (0.0842) Prec 99.219% (96.929%)

Epoch: [146][200/391] Time 0.020 (0.026) Data 0.000 (0.004) Loss
0.0992 (0.0839) Prec 97.656% (97.042%)

Epoch: [146][300/391] Time 0.020 (0.026) Data 0.000 (0.004) Loss
0.0549 (0.0851) Prec 98.438% (97.015%)

Validation starts

Test: [0/79] Time 0.092 (0.092) Loss 0.3052 (0.3052) Prec 91.406%
(91.406%)

* Prec 89.020%

best acc: 89.510000

Epoch: [147][0/391] Time 0.145 (0.145) Data 0.116 (0.116) Loss
0.0756 (0.0756) Prec 96.875% (96.875%)

Epoch: [147][100/391] Time 0.022 (0.027) Data 0.002 (0.005) Loss
0.0951 (0.0904) Prec 96.094% (96.875%)

Epoch: [147][200/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.0605 (0.0906) Prec 97.656% (96.801%)

Epoch: [147][300/391] Time 0.020 (0.026) Data 0.000 (0.005) Loss
0.1291 (0.0873) Prec 94.531% (96.893%)

Validation starts

Test: [0/79] Time 0.092 (0.092) Loss 0.2654 (0.2654) Prec 91.406%
(91.406%)

* Prec 89.020%

best acc: 89.510000

Epoch: [148][0/391] Time 0.158 (0.158) Data 0.121 (0.121) Loss
0.0431 (0.0431) Prec 98.438% (98.438%)

Epoch: [148][100/391] Time 0.022 (0.027) Data 0.002 (0.004) Loss
0.1520 (0.0847) Prec 94.531% (97.092%)

Epoch: [148][200/391] Time 0.028 (0.026) Data 0.008 (0.004) Loss
0.0624 (0.0886) Prec 96.875% (96.964%)

Epoch: [148][300/391] Time 0.026 (0.025) Data 0.007 (0.004) Loss
0.0761 (0.0890) Prec 96.875% (96.919%)

Validation starts

Test: [0/79] Time 0.094 (0.094) Loss 0.3034 (0.3034) Prec 90.625%
(90.625%)


```

* Prec 89.480%
best acc: 89.510000
Epoch: [149][0/391]      Time 0.152 (0.152)      Data 0.115 (0.115)      Loss
0.0418 (0.0418)      Prec 99.219% (99.219%)
Epoch: [149][100/391]    Time 0.028 (0.026)      Data 0.008 (0.006)      Loss
0.0607 (0.0836)      Prec 98.438% (97.084%)
Epoch: [149][200/391]    Time 0.030 (0.026)      Data 0.010 (0.005)      Loss
0.0682 (0.0862)      Prec 97.656% (97.003%)
Epoch: [149][300/391]    Time 0.030 (0.026)      Data 0.011 (0.006)      Loss
0.0585 (0.0869)      Prec 98.438% (96.989%)
Validation starts
Test: [0/79]      Time 0.105 (0.105)      Loss 0.3030 (0.3030)      Prec 89.844%
(89.844%)
* Prec 89.000%
best acc: 89.510000

```

```

[13]: PATH = "result/VGG16_project_2b4b/model_best.pth.tar"
model_name = "VGG16_project_2b4b"
model = VGG16_project()
checkpoint = torch.load(PATH)
model.load_state_dict(checkpoint['state_dict'])
device = torch.device("cuda")

model.cuda()
model.eval()

test_loss = 0
correct = 0

with torch.no_grad():
    for data, target in testloader:
        data, target = data.to(device), target.to(device) # loading to GPU
        output = model(data)
        pred = output.argmax(dim=1, keepdim=True)
        correct += pred.eq(target.view_as(pred)).sum().item()

test_loss /= len(testloader.dataset)

print('\nTest set: Accuracy: {}/{} ({:.0f}%) \n'.format(
    correct, len(testloader.dataset),
    100. * correct / len(testloader.dataset)))

```

Test set: Accuracy: 8951/10000 (90%)

```
[5]: class SaveOutput:
    def __init__(self):
        self.outputs = []
    def __call__(self, module, module_in):
        self.outputs.append(module_in)
    def clear(self):
        self.outputs = []

##### Save inputs from selected layer #####
save_output = SaveOutput()
i = 0
counter = 0
for layer in model.modules():
    i+=1
    if isinstance(layer, QuantConv2d):
        print(i, "-th layer prehooked")
        counter+=1
        print(layer, counter)
        layer.register_forward_pre_hook(save_output)
#####

dataiter = iter(testloader)
images, labels = next(dataiter)
images = images.to(device)
out = model(images)
```

```
3 -th layer prehooked
QuantConv2d(
  3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 1
7 -th layer prehooked
QuantConv2d(
  64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 2
12 -th layer prehooked
QuantConv2d(
  64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 3
16 -th layer prehooked
QuantConv2d(
  128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 4
21 -th layer prehooked
```

```

QuantConv2d(
  128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 5
25 -th layer prehooked
QuantConv2d(
  256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 6
29 -th layer prehooked
QuantConv2d(
  256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 7
34 -th layer prehooked
QuantConv2d(
  256, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 8
38 -th layer prehooked
QuantConv2d(
  16, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 9
41 -th layer prehooked
QuantConv2d(
  16, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 10
46 -th layer prehooked
QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 11
50 -th layer prehooked
QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 12
54 -th layer prehooked
QuantConv2d(
  512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False
  (weight_quant): weight_quantize_fn()
) 13

```

```
[6]: w_bit = 4
```

```

weight_q = model.features[27].weight_q # quantized value is stored during the
↳ training
w_alpha = model.features[27].weight_quant.wgt_alpha # alpha is defined in
↳ your model already. bring it out here
w_delta = w_alpha/(2**(w_bit-1)-1) # delta can be calculated by using alpha
↳ and w_bit
weight_int = weight_q/w_delta # w_int can be calculated by weight_q and w_delta
# print(weight_int) # you should see clean integer numbers
print(f"Weight Int Shape: {weight_int.shape}")

```

Weight Int Shape: torch.Size([16, 16, 3, 3])

```

[7]: x_bit = 2
x = save_output.outputs[8][0] # input of the 2nd conv layer
print(f"Input Shape: {x.shape}")
x_alpha = model.features[27].act_alpha
x_delta = x_alpha/(2**x_bit-1)

act_quant_fn = act_quantization(x_bit) # define the quantization function
x_q = act_quant_fn(x, x_alpha) # create the quantized value for x

x_int = x_q/x_delta
print(f"Input Int Sample: {x_int[0,0,0,:5]}")

```

Input Shape: torch.Size([128, 16, 4, 4])

Input Int Sample: tensor([2.0000, 3.0000, 2.0000, 1.0000], device='cuda:0',
grad_fn=<SelectBackward0>)

```

[8]: ##### input floating number / weight quantized version

conv_ref = torch.nn.Conv2d(in_channels = 16, out_channels=16, kernel_size = 3,
↳ padding=1, bias = False)
conv_ref.weight = torch.nn.parameter.Parameter(weight_int)
output_int = conv_ref(x_int)

output_recovered = output_int*x_delta*w_delta
relu = nn.ReLU(inplace=True)
relu_output_recovered = relu(output_recovered)

output_ref = save_output.outputs[9][0]

```

```

[9]: difference = abs( output_ref - relu_output_recovered )
print(difference.mean()) ## It should be small, e.g., 2.3 in my trained model

```

tensor(1.3916e-06, device='cuda:0', grad_fn=<MeanBackward0>)

```
[11]: x_pad = torch.zeros(128, 16, 6, 6).to(x_int.device)
x_pad[:, :, 1:5, 1:5] = x_int
X = x_pad[0]
X = X.reshape(X.size(0), -1)
X.size()
```

```
[11]: torch.Size([16, 36])
```

```
[ ]: bit_precision = 4
file_path = 'activation.txt'

with open(file_path, 'w') as f:
    f.write('// Format: [Row7_MSB ... Row0_LSB] (Each row is 4-bit)\n')
    f.write('// Time step 0 to 35\n')

    for i in range(X.size(1)):
        line_bin = ""

        for j in range(X.size(0)):
            val = int(round(X[7-j, i].item()))

            val_2s = val & 0xF

            line_bin += f'{val_2s:04b}'

        f.write(line_bin + '\n')
```

```
[ ]: weight_int.size()
```

```
[ ]: torch.Size([8, 8, 3, 3])
```

```
[ ]: W = torch.reshape(weight_int, (weight_int.size(0), weight_int.size(1), -1))
W.size()
```

```
[ ]: torch.Size([8, 8, 9])
```

```
[ ]: W = weight_int.reshape(weight_int.size(0), weight_int.size(1), -1)

bit_precision = 4
file_path = 'weight.txt'

with open(file_path, 'w') as f:
    f.write('// Format: [Row7_MSB ... Row0_LSB] for specific Col and_
↳Kernel_idx\n')
    f.write('// Layout: Loops are K_ij -> Output_Ch(Col) -> Input_Ch(Row)\n')

    for kij in range(W.size(2)):
```

```

    for i in range(W.size(0)):
        line_bin = ""

        for j in range(W.size(1)):
            val = int(round(W[i, 7-j, kij].item()))

            val_2s = val & 0xF

            line_bin += f'{val_2s:04b}'

        f.write(line_bin + '\n')

```

```
[ ]: W[0,:,0]
```

```
[ ]: tensor([-5.0000, -0.0000,  2.0000,  7.0000,  7.0000,  3.0000, -3.0000,  2.0000],
           device='cuda:0', grad_fn=<SelectBackward0>)
```

```

[ ]: psum = torch.zeros(8, X.size(1), 9).to(device)

for kij in range(9):
    W_k = W[:, :, kij]

    psum_slice = torch.matmul(W_k.float(), X.float())

    psum[:, :, kij] = psum_slice

bit_precision = 16
file_path = 'psum.txt'

with open(file_path, 'w') as f:
    f.write('// Format: [Col7_MSB ... Col0_LSB] (Each 16-bit)\n')
    f.write('// Order: Loop Kernel(9) -> Loop Time(36)\n')

    # Loop 1: Kernel Index (0-8)
    for kij in range(9):

        for i in range(psum.size(1)):
            line_bin = ""

            for j in range(psum.size(0)):
                val = int(round(psum[7-j, i, kij].item()))

                val_2s = val & 0xFFFF

                line_bin += f'{val_2s:016b}'

```

```
f.write(line_bin + '\n')
```

```
[ ]: import torch
import torch.nn.functional as F

out_relu = F.relu(output_int)
out = out_relu[0]
out = out.reshape(out.size(0), -1)

bit_precision = 16
file_path = 'output.txt'

with open(file_path, 'w') as f:
    f.write('// Format: [Col7_MSB ... Col0_LSB] (Each 16-bit)\n')
    f.write('// Final Output after ReLU\n')

    # Loop Time (Input Sequence)
    for i in range(out.size(1)):
        line_bin = ""

        # Loop Channel (Row 7 -> Row 0)
        for j in range(out.size(0)):
            val = int(round(out[7-j, i].item()))

            val_bin = val & 0xFFFF

            line_bin += f'{val_bin:016b}'

        f.write(line_bin + '\n')
```

output.txt generated with 16 lines.